

第二章 控制系统的数学模型

物理性质不同的系统可以有相同的数学模型

精确 + 简明

2.1 控制系统微分方程的建立

$$C^{(n)}(t) + a_1 C^{(n-1)}(t) + \dots + a_n C(t) =$$

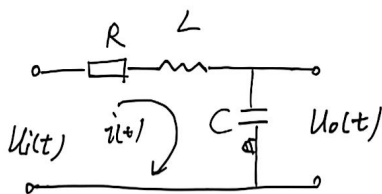
$$b_0 r^{(m)}(t) + b_1 r^{(m-1)}(t) + \dots + b_n r(t)$$

$n \geq m$ n 阶系统

$C(t)$ 输出

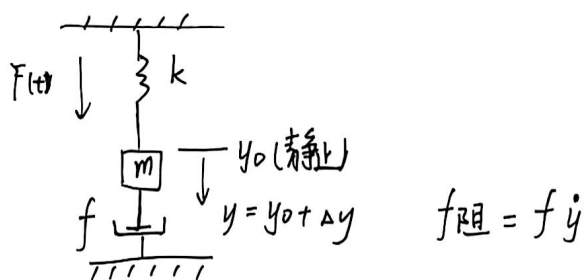
$r(t)$ 输入

一、电气系统



$$\frac{d^2 u_o(t)}{dt^2} + \frac{R}{L} \frac{du_o(t)}{dt} + \frac{1}{LC} u_o(t) = \frac{1}{LC} u_i(t)$$

二、机械系统



$$\ddot{\Delta y} + \frac{f}{m} \dot{\Delta y} + \frac{k}{m} \Delta y = \frac{1}{m} F(t)$$

2.2 传递函数

零初始条件 $r^{(i)}(0) = 0, i = 0, 1, \dots, m-1$

$C^{(j)}(0) = 0, j = 0, 1, \dots, n-1$

(3). 定理

① 线性定理

$$\mathcal{L}[k_1 f_1(t) + k_2 f_2(t)] = k_1 \mathcal{L}[f_1(t)] + k_2 \mathcal{L}[f_2(t)]$$

② 微分定理

$$\text{初始为零 } \mathcal{L}[f^{(n)}(t)] = s^n \mathcal{L}[f(t)]$$

③ 终值定理

$$\lim_{t \rightarrow \infty} f(t) = \lim_{s \rightarrow 0} s \cdot F(s)$$

1. (1) 拉氏变换

$$F(s) = \mathcal{L}[f(t)] = \int_0^{\infty} f(t) e^{-st} dt$$

(2) 常用

$$\mathcal{L}[\delta(t)] = 1 \quad \mathcal{L}[\sin \omega t] = \frac{\omega}{s^2 + \omega^2}$$

$$\mathcal{L}[t] = \frac{1}{s^2} \quad \mathcal{L}[\cos \omega t] = \frac{s}{\omega^2 + s^2}$$

$$\mathcal{L}[u(t)] = \frac{1}{s}$$

$$\mathcal{L}[t^n] = \frac{n!}{s^{n+1}}$$



2. 定义

初始条件为零时, $G(s) = \frac{C(s)}{R(s)} = \frac{b_0 s^m + \dots + b_m}{s^n + a_1 s^{n-1} + \dots + a_n}$

$$G(s) = \frac{N(s)}{D(s)} = k \frac{(s-z_1)(s-z_2)\dots(s-z_m)}{(s-p_1)(s-p_2)\dots(s-p_n)}$$

有理分式

$z_1, \dots, z_m$ 零点

$p_1, \dots, p_n$ 极点

有限

$n > m$, 有 $n-m$ 个无穷零点

闭环传递函数 $\phi(s) = \frac{*}{(s-p_1)(s-p_2)\dots(s-p_n)}$ 闭环极点 $p_1, \dots, p_n$

$$= \frac{N(s)}{D(s)}$$

$D(s)$ 特征多项式

$D(s)=0$ 特征方程

$\lambda_1, \dots, \lambda_n$ 特征根

3. 基本环节及其传递函数

1) 放大环节

$$C(t) = K \cdot r(t)$$

$$G(s) = K$$

(5) 纯微分环节

$$C(t) = \frac{dr(t)}{dt}$$

$$G(s) = s$$

2) 惯性环节

$$T \frac{dC(t)}{dt} + C(t) = r(t)$$

$$G(s) = \frac{1}{Ts+1}$$

(6) 一阶微分环节

$$G(t) = T \frac{dr(t)}{dt} + r(t)$$

$$G(s) = Ts+1$$

3) 积分环节

$$C(t) = \int r(t) dt$$

$$G(s) = \frac{1}{s}$$

可用以输出信号不变

(7) 二阶微分环节

$$T^2 \frac{d^2 r(t)}{dt^2} + 2\zeta T \frac{dr(t)}{dt} + r(t) = C(t)$$

$$G(s) = T^2 s^2 + 2\zeta Ts + 1$$

(4) 振荡环节

$$T^2 \frac{d^2 C(t)}{dt^2} + 2\zeta T \frac{dC(t)}{dt} + C(t) = r(t)$$

时间常数

$0 < \zeta < 1$ 阻尼比

$$G(s) = \frac{1}{T^2 s^2 + 2\zeta Ts + 1} = \frac{\omega_n^2}{s^2 + 2\zeta \omega_n s + \omega_n^2}$$

$\omega_n = \frac{1}{T}$ 无阻尼自然频率 rad/s

(8) 延迟环节

$$C(t) = r(t-T)$$

T : 延迟时间

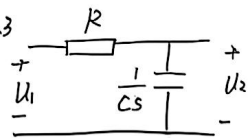
$$G(s) = e^{-Ts}$$



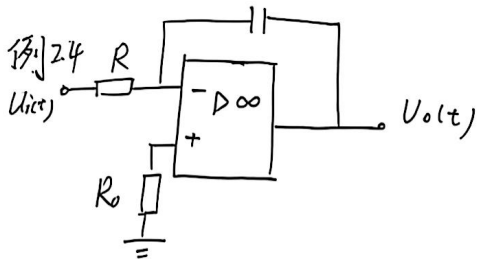
4 电路网络运算阻抗

$$R \rightarrow R, L \rightarrow sL, C \rightarrow \frac{1}{sC}$$

例23



$$\frac{U_2(s)}{U_1(s)} = \frac{\frac{1}{sC}}{R + \frac{1}{sC}} = \frac{1}{RCs + 1}$$



$$\frac{U_1(s)}{R} + \frac{U_0(s)}{\frac{1}{sC}} = 0$$

$$\frac{U_0(s)}{U_1(s)} = -\frac{1}{RCs}$$

2.3 控制系统的框图和传递函数

一、概念

1. 定义

2. 组成

(1) 函数方框: $\boxed{\frac{1}{s}}$, $\boxed{G(s)}$

(2) 信号流线: $\longrightarrow$

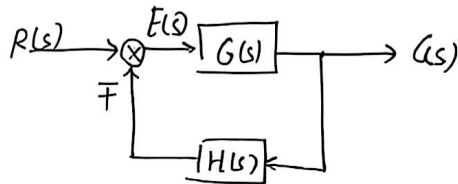
单向, 不可逆

(3) 表示各种变量的字母

(4) 相加点: $U_3 = U_1 - U_2$

(5) 分支点: U_1

偏差 (反馈为1, 等于误差)



前向通道传递函数 $G(s)$

开环传递函数 $G(s) \cdot H(s)$

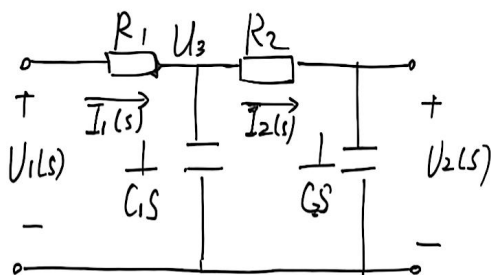
$$C(s) = G(s) \cdot E(s)$$

$$= G(s) [R(s) - H(s) C(s)]$$

$$[1 \pm G(s) H(s)] C(s) = G(s) R(s)$$

$$\phi(s) = \frac{G(s)}{1 \pm G(s) H(s)} \text{ 闭环传递函数}$$

例2.5

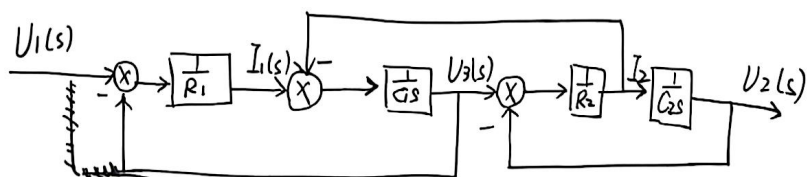


$$U_2(s) = \frac{1}{C_2 s} \cdot I_2(s)$$

$$I_2(s) = \frac{1}{R_2} [U_3(s) - U_2(s)]$$

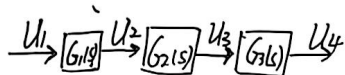
$$U_3(s) = \frac{1}{C_1 s} [I_1(s) - I_2(s)]$$

$$I_1(s) = \frac{1}{R_1} [U_1(s) - U_3(s)]$$



二. 框图的变换规则 “等效”

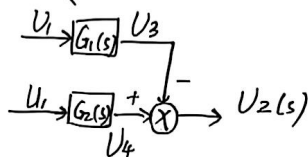
1. 串联 “积”



$$\begin{aligned} U_4 &= G_3(s) \cdot U_3 \\ &= G_3(s) \cdot G_2(s) \cdot U_2 \\ &= G_3(s) \cdot G_2(s) \cdot G_1(s) \cdot U_1 \end{aligned}$$

$$G(s) = G_1(s) \cdot G_2(s) \cdot G_3(s)$$

2. 并联 “代数和”



$$\begin{aligned} U_2(s) &= U_4(s) - U_3(s) \\ &= G_2(s) \cdot U_1 - G_1(s) \cdot U_1 \end{aligned}$$

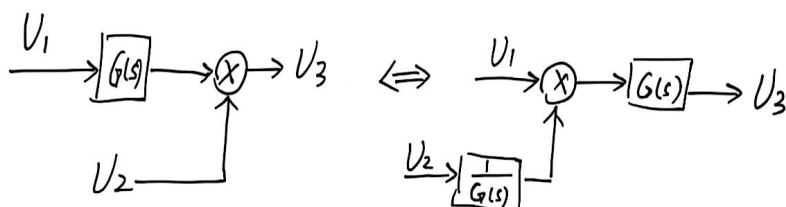
$$G(s) = -G_1(s) + G_2(s)$$

3. 反馈

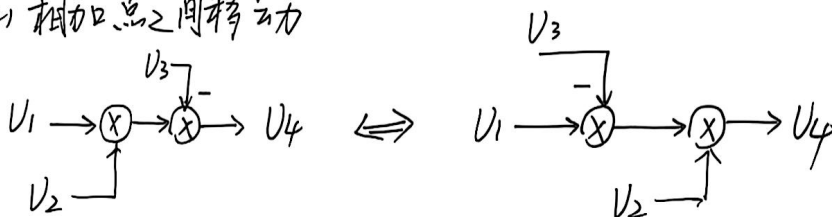
$$\phi(s) = \frac{G(s)}{1 \pm G(s)H(s)}$$

4. 相加点和分支点移动

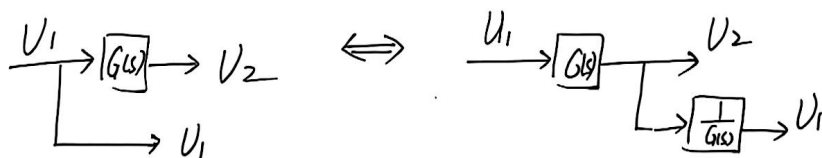
(1) 相加点前移



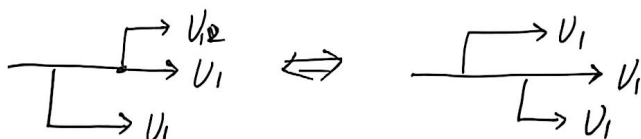
(2) 相加点之间移动



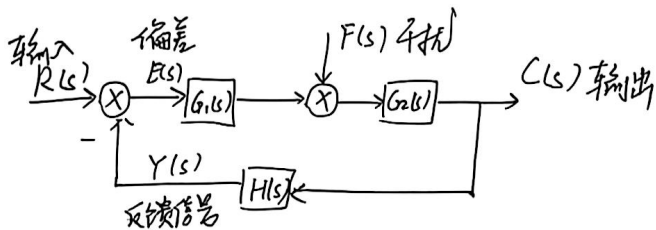
(3) 分支点后移



(4) 相邻分支点间移动



三、闭环系统的传递函数



1. 开环传递函数

$$G(s) = G_1(s) G_2(s) H(s)$$

2. 闭环传递函数 $R(s) \rightarrow C(s)$

$$\phi(s) = \frac{C(s)}{R(s)} = \frac{G_1(s) G_2(s)}{1 + G_1(s) G_2(s) H(s)}$$

3. $F(s) \rightarrow C(s)$

$$\phi_F(s) = \frac{C(s)}{F(s)} = \frac{G_2(s)}{1 + G_1(s) G_2(s) H(s)}$$

4. 总输出

$$\bar{C}(s) = \frac{G_1 G_2}{1 + G_1 G_2 H} R(s) + \frac{G_2}{1 + G_1 G_2 H} F(s)$$

5. $R(s) \rightarrow E(s)$

$$\phi_{ER}(s) = \frac{1}{1 + G_1 G_2 H}$$

6. $F(s) \rightarrow E(s)$

$$\phi_{EF}(s) = \frac{-G_2 H}{1 + G_1 G_2 H}$$

7. 总偏差

$$E(s) = \frac{1}{1 + G_1 G_2 H} R(s) - \frac{G_2 H}{1 + G_1 G_2 H} F(s)$$

四、化简

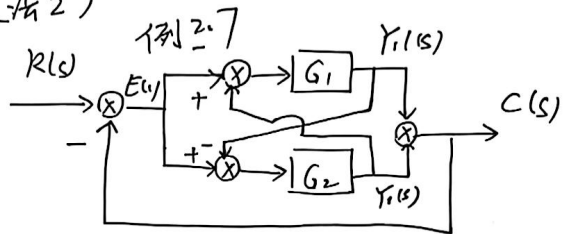
(法1) 前移、后移 (框图化简)

例27法3: 4条前向、5个回路 $n=4$

$$L_1 = G_1 \quad L_2 = -G_2 \quad L_3 = -G_1 G_2 \quad L_4 = -G_1 G_2 \quad L_5 = -G_1 G_2$$

$$\Delta = 1 - G_1 + G_2 + 3G_1 G_2$$

(法2)



$$\begin{cases} Y_1 = G_1 [-E + Y_2] \\ Y_2 = G_2 [E - Y_1] \end{cases}$$

$$\Rightarrow \phi_1 = \frac{G_1 G_2 - G_1}{G_1 G_2 + 1}$$

$$\phi_2 = \frac{G_1 G_2 + G_2}{G_1 G_2 + 1}$$

$$\phi = \phi_1 + \phi_2$$

$$\phi(s) = \frac{\phi}{1 + \phi}$$

(法3) 梅森增益公式

$$\phi(s) = \frac{\sum_{k=1}^n P_k \Delta_k}{\Delta}$$

$$\text{特征式 } \Delta = 1 - \sum L_i + \sum L_i L_j - \sum L_i L_j L_k + \dots$$

L_i : 回路传递函数 (有正负号)

L_i, L_j : 互不接触的两条回路传递函数之积

n : 前向通路个数

P_k : 从输入到输出第 k 条前向通路上各传递函数之积

Δ_k : 余因子, Δ 中与第 k 条所在项除去后余下的前向通路接触回路

$$P_1 = -G_1 \quad \Delta_k = 1, \quad k=1, 2, 3, 4$$

$$P_2 = G_2$$

$$P_3 = G_1 G_2$$

$$P_4 = G_1 G_2$$

$$\phi(s) = \frac{-G_1 + G_2 + 2G_1 G_2}{1 - G_1 + G_2 + 3G_1 G_2}$$



2.4 非线性系统的线性化

一、线性系统性质

1. 叠加性

$$\text{若 } u_1(t) \rightarrow y_1(t)$$

$$u_2(t) \rightarrow y_2(t)$$

$$\text{则 } u_1(t) + u_2(t) \rightarrow y_1(t) + y_2(t)$$

2. 齐次性

$$u_1(t) \rightarrow y_1(t)$$

$$k \cdot u_1(t) \rightarrow k \cdot y_1(t)$$

$$\text{例 1. } y = x^2 \quad x$$

$$\text{例 2. } y = mx + k \quad x \text{ 增量线性}$$

二、小偏差线性化

$$y = f(x)$$

$$y = f(x_0) + \left. \frac{df(x)}{dx} \right|_{x=x_0} \Delta x + o(\Delta x)$$

$$\Delta y = \left. \frac{df(x)}{dx} \right|_{x=x_0} \Delta x$$

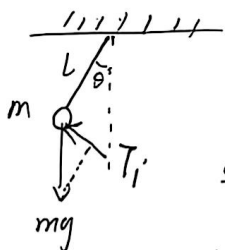
$$y = f(x, \dot{x}, \ddot{x}) \quad (\text{点 } 10, 0, 0, 0)$$

$$y = y_0 + \left. \frac{\partial f}{\partial x} \right|_0 \Delta x + \left. \frac{\partial f}{\partial \dot{x}} \right|_0 \Delta \dot{x} + \left. \frac{\partial f}{\partial \ddot{x}} \right|_0 \Delta \ddot{x} + o(\quad)$$

$$\text{例 3. } y = \ddot{x} + \frac{1}{2} \dot{x} + 2x + x^2 \text{ 在原点附近线性化}$$

$$\left. \frac{\partial y}{\partial x} \right|_0 = 2x + 2 \Big|_{x=0} = 2 \quad \frac{\partial y}{\partial \dot{x}} = \frac{1}{2} \quad \frac{\partial y}{\partial \ddot{x}} = 1$$

$$y = 2x + \frac{1}{2} \dot{x} + \ddot{x}$$



$$T_1 - mg \cos \theta = m l \ddot{\theta}$$

$$\sin \theta = \theta - \frac{1}{3!} \theta^3 + \dots \approx \theta$$

$$m l \ddot{\theta} + m g l \theta = T_1$$



第三章 线性系统的时域分析与综合

3.1 引言

一、典型输入信号

1. 阶跃信号

$$r(t) = \begin{cases} R \cdot u(t) & t \geq 0 \\ 0 & t < 0 \end{cases}$$

$$R(s) = \frac{R}{s} \quad R=1 \text{ 单位}$$

2. 斜坡信号

$$r(t) = \begin{cases} R \cdot t & t \geq 0 \\ 0 & t < 0 \end{cases}$$

$$R(s) = \frac{R}{s^2} \quad R=1 \text{ 单位}$$

3. 加速信号

$$r(t) = \begin{cases} R \cdot t^2 & t \geq 0 \\ 0 & t < 0 \end{cases}$$

$$R(s) = \frac{2R}{s^3} \quad R=\frac{1}{2} \text{ 单位}$$

4. 脉冲信号 (函数)

$$\delta_h(t) = \begin{cases} \frac{1}{h}, & 0 \leq t \leq h \\ 0, & \text{其他} \end{cases}$$

$$R(s) =$$

5. 正弦信号

$$r(t) = A \sin \omega t$$

幅值/振幅 角频率

$$R(s) = \frac{A\omega}{s^2 + \omega^2}$$

二、单位冲激响应

$$r(t) = \delta(t), \quad R(s) = 1$$

$$C(s) = G(s) R(s) = G(s)$$

$$\mathcal{L}^{-1}: C(t) = \mathcal{L}^{-1}[G(s)] = g(t)$$

$$\text{卷积定理} \quad c(t) = g(t) * r(t)$$

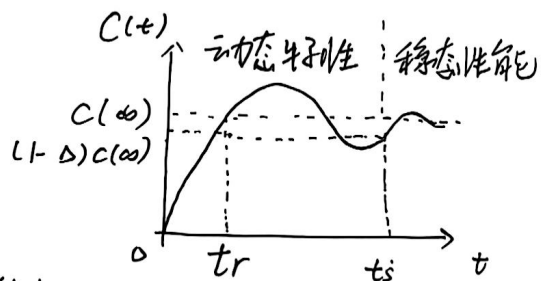
$$= \int_0^\infty g(\tau) r(t-\tau) d\tau$$

$$= \int_0^\infty g(t-\tau) r(\tau) d\tau$$

$$C(t) = \underbrace{\text{稳态响应}}_{r(t)} + \underbrace{\text{瞬态响应}}_{G(s)}$$

$$C(s) = G(s) \cdot R(s)$$

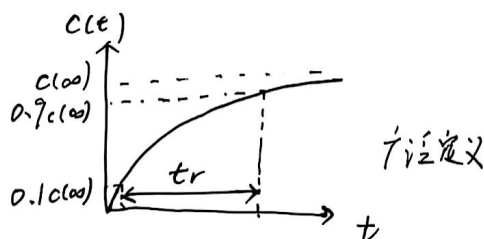
三、时间响应的性能指标



h : 脉冲宽度
冲激 $\delta(t) = \lim_{h \rightarrow 0} \delta_h(t)$

$\downarrow \mathcal{L}$
1

1. 上升时间 $C(t_r) = C(\infty)$



2. 峰值时间 t_p

$$\left. \frac{dC(t)}{dt} \right|_{t=t_p} = 0$$



3. 最大超调量

$$\sigma_p = \frac{C(tp) - C(\infty)}{C(\infty)} \times 100\%$$

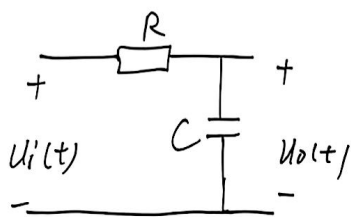
4. 过渡过程时间 t_s

$$\Delta = 0.02 \text{ 或 } 0.05$$

5. 振荡次数 N - 峰-谷

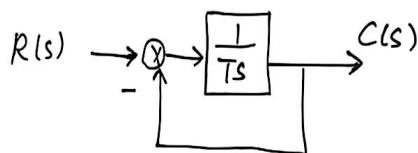
$$N=1$$

3.2 一阶系统的时域分析



$$G(s) = \frac{1}{Ts+1}$$

$$T = RC$$



一. 单位阶跃响应

$$r(t) = u(t), \quad R(s) = \frac{1}{s}$$

$$C(s) = G(s) \cdot R(s)$$

$$= \frac{1}{Ts+1} \cdot \frac{1}{s}$$

$$= \frac{1}{s} - \frac{T}{Ts+1}$$

$$L^{-1}: C(t) = 1 - e^{-\frac{1}{T}t} \quad (t \geq 0)$$

$$C(0) = 0$$

$$C(T) = 0.632$$

$$C(2T) = 0.865$$

$$C(3T) = 0.95$$

$$C(4T) = 0.982$$

$$t_s = (3 \sim 4)T$$

$$\left. \frac{dC(t)}{dt} \right|_{t=0} = \frac{1}{T} e^{-\frac{t}{T}} = \frac{1}{T}$$

响应速度 $\uparrow \rightarrow t_s \downarrow \rightarrow T \downarrow$

闭环极点与
虚轴越远

响应速度: t_r, t_p, t_s

~~阻尼系数~~

阻尼程度: σ_p, N, t_s

二. 单位斜坡响应

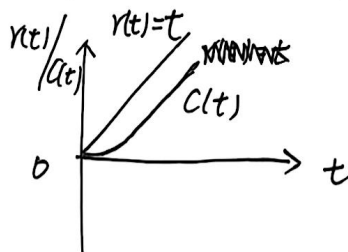
$$r(t) = t, \quad R(s) = \frac{1}{s^2}$$

$$C(s) = G(s) \cdot R(s)$$

$$= \frac{1}{Ts+1} \cdot \frac{1}{s^2}$$

$$= \frac{1}{s^2} - \frac{T}{s} + \frac{T}{Ts+1}$$

$$L^{-1}: C(t) = t - T + T e^{-\frac{1}{T}t}, \quad t \geq 0$$



三. 单位冲激响应

$$r(t) = \delta(t), \quad R(s) = 1$$

$$C(t) = \frac{1}{T} e^{-\frac{t}{T}}$$



3.3 二阶系统的时域分析

一、典型形式

$$R(s) \rightarrow \boxed{\phi(s)} \rightarrow C(s)$$

$$\frac{d^2 c(t)}{dt^2} + 2\zeta \omega_n \frac{dc(t)}{dt} + \omega_n^2 c(t) = \omega_n^2 r(t)$$

ζ : 阻尼比

ω_n : 无阻尼自然振荡频率

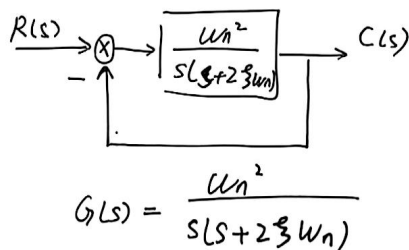
$$\phi(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

$$G'(s) = \frac{k}{s(Ts+1)}$$

$$\begin{cases} \frac{1}{T} = 2\zeta\omega_n \\ k = \frac{\omega_n^2}{2\zeta} \end{cases}$$

$$\phi'(s) = \frac{k}{Ts^2 + s + k}$$

$$\begin{aligned} \omega_n &= \sqrt{\frac{k}{T}} \\ \zeta &= \frac{1}{2\sqrt{TK}} \end{aligned}$$



$$s^2 + 2\zeta\omega_n s + \omega_n^2 = 0$$

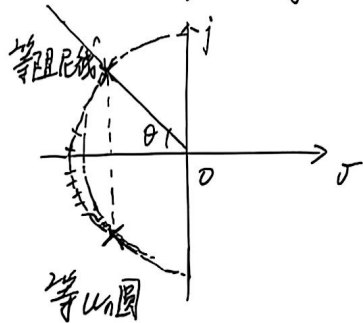
$$s_{1,2} = -\zeta\omega_n \pm \omega_n \sqrt{\zeta^2 - 1}$$

4. $\zeta > 1$ 过阻尼

$$s_{1,2} = -\zeta\omega_n \pm \omega_n \sqrt{\zeta^2 - 1}$$

1. 欠阻尼 $0 < \zeta < 1$

$$s_{1,2} = -\zeta\omega_n \pm j\omega_n \sqrt{1 - \zeta^2}$$



$$\theta = \tan^{-1} \frac{\sqrt{1 - \zeta^2}}{\zeta}$$

ζ 越小, θ 越大

5. $-1 < \zeta < 0$ 负阻尼

$$s_{1,2} = -\zeta\omega_n \pm j\omega_n \sqrt{1 - \zeta^2}$$

2. $\zeta = 0$ 无阻尼

$$s_{1,2} = \pm j\omega_n$$

3. $\zeta = 1$ 临界阻尼

$$s_{1,2} = -\omega_n$$



二、单位阶跃响应

$$r(t) = u(t), \quad R(s) = \frac{1}{s}$$

$$C(s) = \phi(s) \cdot R(s)$$

$$= \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} \cdot \frac{1}{s} = \frac{1}{s} - \frac{s + 2\zeta\omega_n}{(s + \zeta\omega_n + j\omega_d)(s + \zeta\omega_n - j\omega_d)}$$

($\omega_d = \omega_n \sqrt{1-\zeta^2}$ 有阻尼自然振荡频率)

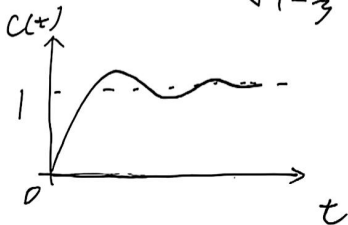
$$= \frac{1}{s} - \frac{s + \zeta\omega_n}{(s + \zeta\omega_n)^2 + \omega_d^2} - \frac{\zeta\omega_n}{(s + \zeta\omega_n)^2 + \omega_d^2} \cdot \frac{\omega_d}{\omega_d}$$

$$\mathcal{L}^{-1}: c(t) = 1 - e^{-\zeta\omega_n t} \cos\omega_d t - \frac{\zeta\omega_n}{\omega_d} \cdot e^{-\zeta\omega_n t} \sin\omega_d t$$

$$= 1 - e^{-\zeta\omega_n t} \left[\cos\omega_d t + \frac{\zeta}{\sqrt{1-\zeta^2}} \sin\omega_d t \right]$$

$$A = \frac{1}{\sqrt{1-\zeta^2}}, \quad \phi = \tan^{-1} \frac{\sqrt{1-\zeta^2}}{\zeta}$$

$$c(t) = 1 - \frac{e^{-\zeta\omega_n t}}{\sqrt{1-\zeta^2}} \sin(\omega_d t + \phi) \quad \text{稳定} \Leftrightarrow \text{闭环极点位于左半平面}$$



$$T_d = \frac{2\pi}{\omega_d}$$

3. 最大超调量

$$\sigma_p = \frac{c(t_p) - c(\infty)}{c(\infty)} \times 100\% \\ = e^{-\frac{\zeta\pi}{\sqrt{1-\zeta^2}}} \times 100\%$$

三、欠阻尼系统的动态性能指标

1. 上升时间 t_r

$$c(t_r) = 1$$

$$t_r = \frac{\pi - \phi}{\omega_d}$$

2. 峰值时间 t_p

$$\frac{dc(t)}{dt} = 0$$

$$t_p = \frac{\pi}{\omega_d} = \frac{1}{2} T_d$$

4. 过渡过程时间

$$t_s = \frac{3 \sim 4}{\zeta\omega_n}$$

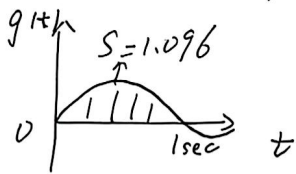
5. 振荡次数

$$N = \frac{t_s}{T_d} = \begin{cases} \frac{1.5\sqrt{1-\zeta^2}}{\pi\zeta} & \Delta = 0.05 \\ \frac{2\sqrt{1-\zeta^2}}{\pi\zeta} & \Delta = 0.02 \end{cases}$$



四、二阶系统单位冲激响应

$$C(t) = 1 - \frac{e^{-\zeta \omega_n t}}{\sqrt{1-\zeta^2}} \sin(\omega_d t + \phi)$$



$$t_p = 1 \text{ sec}$$

$$C(t_p) = 1.096$$

$$\sigma_p = 9.6\%$$

五、初始条件不为零时二阶系统时间响应

$$\mathcal{L}[f^{(n)}] = s^n F(s) - s^{n-1} f(0) - s^{n-2} \dot{f}(0) - \dots - f^{(n-1)}(0)$$

$$\ddot{C}(t) + 2\zeta \omega_n \dot{C}(t) + \omega_n^2 C(t) = \omega_n^2 r(t)$$

$$[s^2 C(s) - s C(0) - \dot{C}(0)] + 2\zeta \omega_n [s C(s) - C(0)] + \omega_n^2 C(s) = \omega_n^2 R(s)$$

$$(s^2 + 2\zeta \omega_n s + \omega_n^2) C(s) - [s C(0) + \dot{C}(0)] = \omega_n^2 R(s)$$

$$(s^2 + 2\zeta \omega_n s + \omega_n^2) C(s) = \omega_n^2 R(s) + s C(0) + \dot{C}(0)$$

$$C(s) = \frac{\omega_n^2}{s^2 + 2\zeta \omega_n s + \omega_n^2} R(s) + \frac{(s + 2\zeta \omega_n) C(0) + \dot{C}(0)}{s^2 + 2\zeta \omega_n s + \omega_n^2}$$

非零初始条件响应

例1. $C(t) = 1 + e^{-t} - e^{-2t} \quad (t \geq 0)$

$$C(s) = \frac{1}{s} - \frac{2}{s+1} + \frac{1}{s+2}$$

解: $C(0) = 1$, $\left. \frac{dC(t)}{dt} \right|_{t=0} = (-e^{-t} + 2e^{-2t})|_{t=0} = 1$

$$R(s) = \frac{1}{s}$$

设零初始条件下, 单位阶跃响应

$$G(s) = \frac{C(s)}{R(s)} = \frac{2}{s^2 + 3s + 2}$$

$$C(t) = 1 + A_1 e^{-t} + A_2 e^{-2t}$$

$$\frac{dC(t)}{dt} = -A_1 e^{-t} + 2A_2 e^{-2t}$$

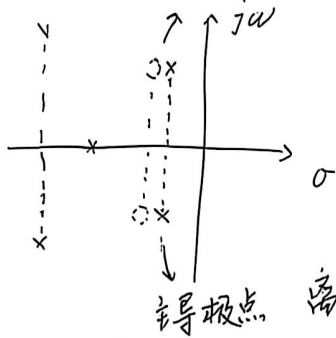
$$C(0) = 0, \quad \dot{C}(0) = 0$$

$$\begin{cases} 1 + A_1 + A_2 = 0 \\ -A_1 - 2A_2 = 0 \end{cases} \quad \begin{cases} A_1 = -2 \\ A_2 = 1 \end{cases}$$

$$C(t) = 1 - 2e^{-t} + e^{-2t}$$



1.4 高阶系统的时域响应根轨迹

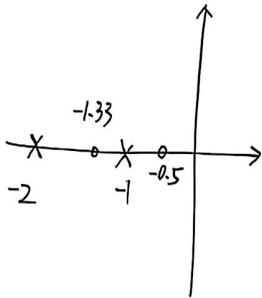


主导极点 离虚轴最近, 其他极点很远, 附近无零点

$$G_1(s) = \frac{4s+2}{(s+1)(s+2)} \quad G_2(s) = \frac{1.5s+2}{(s+1)(s+2)}$$

单位阶跃响应 $G_1(t) = 1 + 2e^{-t} - 3e^{-2t}$

$$G_2(t) = 1 - 0.5e^{-t} - 0.5e^{-2t}$$



3.5 控制系统的稳定性

线性定常系统稳定的充要条件: 所有闭环极点实部小于0

一. 劳斯 (Routh) 稳定判据

$$1. D(s) = a_0 s^n + a_1 s^{n-1} + \dots + a_n = 0$$

稳定的必要条件: $a_i > 0, i=0, 1, \dots, n$

正值不缺项

2. Routh 表

s^n	a_0	a_2	a_4	a_6	...
s^{n-1}	a_1	a_3	a_5	a_7	...
s^{n-2}	b_1	b_2			
s^{n-3}	c_1	c_2			
$\vdots$	$\vdots$	$\vdots$			
s^0					

$$b_1 = \frac{a_1 a_2 - a_0 a_3}{a_1}$$

$$b_2 = \frac{a_1 a_4 - a_0 a_5}{a_1}$$

$$c_1 = \frac{a_3 b_1 - a_1 b_2}{b_1}$$

$$c_2 = \frac{a_5 b_2 - a_3 b_3}{b_1}$$



3. Routh 稳定判据

* 稳定 $\Leftrightarrow$ Routh 表中第一列全都大于 0

不稳定闭环极点个数 = Routh 表第一列符号改变的次数
(实部为正)

例 1. $D(s) = s^4 + 2s^3 + 3s^2 + 4s + 5$

解: Routh 表

s^4	1	3	5
s^3	2	4	0
s^2	1	5	
s^1	-6	0	
s^0	5		

不稳定闭环极点个数为 2

例 2. $D(s) = s^4 + 2s^3 + 3s^2 + 6s + 1 = 0$

法 3: $x = \frac{1}{s}$

解: Routh 表

s^4	1	3	1	不稳定: 2
s^3	2	6	0	
s^2	$0 \rightarrow \varepsilon^+$			
s^1	$\frac{6\varepsilon^+ - 2}{\varepsilon^+} \rightarrow \infty$			
s^0	1			

$D'(x) = x^4 + 6x^3 + 3x^2 + 2x + 1$

x^4	1	3	1	$\frac{16}{3} - \frac{18}{3} = \frac{2}{3}$
x^3	6	2	0	$-\frac{2}{3}$
x^2	$\frac{8}{3}$	1		$\frac{8}{3}$
x^1	$-\frac{1}{4}$	0	不稳定: 2	
x^0	1			

法 1: $0 \rightarrow \varepsilon^+$

法 2: $D'(s) = (s+a) D(s)$ a 为任意实数

两者判定结果一样

$D'(s) = (s+1) D(s) = s^5 + 3s^4 + 5s^3 + 9s^2 + 7s + 1$

s^5	1	5	7	
s^4	3	9	1	
s^3	2	$\frac{20}{3}$	0	不稳定: 2
s^2	-1	1		
s^1	$\frac{26}{3}$	0		
s^0	1			



例3 $D(s) = s^3 + 2s^2 + s + 2 = 0$

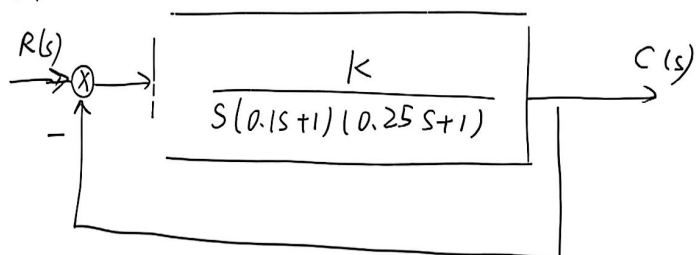
s^3	1	1	s^3	1	1
s^2	2	2	s^2	2	2
s^1	0	0	s^1	4	0
s^0			s^0	2	0

辅助方程: $2s^2 + 2 = 0$ (上一行)

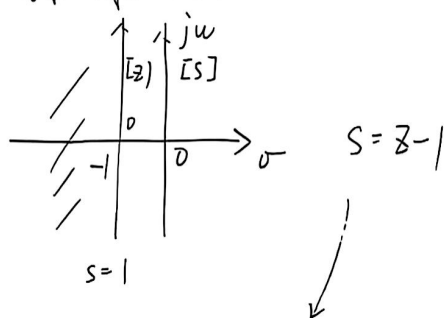
求导: $4s = 0$

用求导后系数代替

例4



要求: 闭环极点全位于 $s = -1$ 垂线之左, 确定 K 取值范围



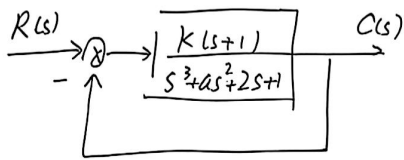
$$D(s) = s(0.1s+1)(0.25s+1) + K$$

$$D(z) = z^3 + 1.1z^2 + 1.5z + 1.4K - 2$$

$$\Rightarrow 0.675 < K < 4.8$$



例5 若系统以 $\omega_n = 2 \text{ rad/s}$ 的角频率作等幅振荡, 确定 k 与 a 的值



$$\text{代入} \begin{cases} -8j - 4a + (k+2) \cdot 2j + k+1 = 0 \\ 8j - 4a - (k+2) \cdot 2j + k+1 = 0 \end{cases}$$

$$\begin{cases} k=2 \\ a=\frac{3}{4} \end{cases}$$

法一: $D(s) = s^3 + as^2 + 2s + 1 + k(s+1)$

$\zeta = 0, \omega_n = 2 \text{ rad/s}$

$s_{1,2} = \pm 2j$

例6: $D(s) = s^5 + 2s^4 + 3s^3 + 6s^2 - 4s - 8 = 0$

$= (s+1)(s-1)(s+2)(s^2+4) = 0$

$s = 1, -1, -2, 2j, -2j$

$s^5 \quad 1 \quad 3 \quad -4$

$s^4 \quad 2 \quad 6 \quad -8$

s^3

s^2

s^1

s^0

辅助方程 $2s^4 + 6s^2 - 8 = 0$

$D(s) = (2s^4 + 6s^2 - 8)(0.5s + 1)$

$= 2(s^2 + 4)(s^2 - 1)(0.5s + 1)$

$= (s^2 + 4)(s^2 - 1)(s + 2) = 0$

法二: Routh 表 $s^3 \quad 1 \quad k+2$

$s^2 \quad a \quad k+1$

$s^1 \quad \frac{a(k+2)-(k+1)}{a} \quad 0$

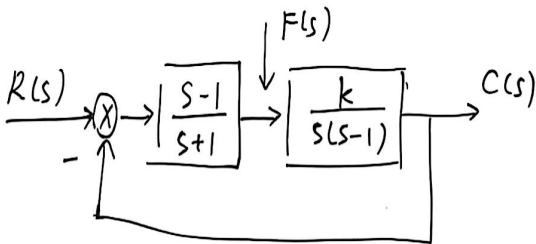
$s^0 \quad k+1$

等幅振荡, s^1 行全为 0

$\begin{cases} \frac{a(k+2)-(k+1)}{a} = 0 \end{cases}$

辅助方程: $\begin{cases} as^2 + k+1 = 0 \quad (s^2+4=0) \end{cases}$

例7. 试分析系统的稳定性



左半平面可相消

右半平面不可相消

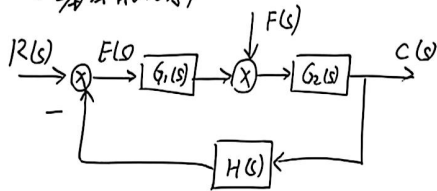
$$\frac{C(s)}{F(s)} = \frac{\frac{k}{s(s-1)}}{1 + \frac{s-1}{s+1} \cdot \frac{k}{s(s-1)}} = \frac{k(s+1)}{(s-1)[s^2+s+k(s-1)]}$$

$s-1$ 不能相消



3. 控制系统的稳态误差

一、基本概念



1. 希望输出

$$E(s) = 0 \text{ 时}$$

$$R(s) - H(s)G(s) = 0$$

$$C_r(s) = \frac{1}{H(s)} \cdot R(s)$$

2. 误差

$$e_1(t) = C_r(t) - C(t)$$

$$= \text{瞬态分量} + \text{稳态分量}$$

3. 稳态误差

$$e_{1ss}(\infty) = \lim_{t \rightarrow \infty} e_1(t)$$

$$= \lim_{s \rightarrow 0} s E_1(s)$$

4. 误差与偏差

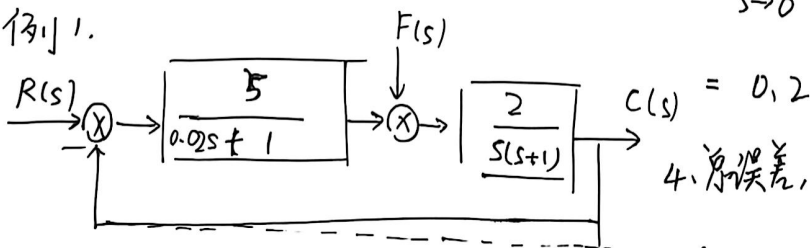
$$E_1(s) = C_r(s) - C(s)$$

$$= \frac{1}{H(s)} \cdot R(s) - C(s)$$

$$= \frac{1}{H(s)} [R(s) - H(s)C(s)]$$

$$\therefore E_1(s) = \frac{1}{H(s)} \cdot E(s)$$

例 1.



已知 $r(t) = t$, $f(t) = -u(t)$

求稳态误差

解: 1. 判断

$$D(s) = 0.02s^3 + 1.02s^2 + s + 10 = 0$$

Routh 表

s^3	0.02	1
s^2	1.02	10
s^1	$\frac{0.82}{1.02}$	0
s^0	10	

$\therefore$ Routh 表中第一列元素都为正

$\therefore$ 闭环系统稳定

2. $r(t) = t$ 单独作用时

$$R(s) = \frac{1}{s^2}$$

$$E_r(s) = \frac{1}{1 + \frac{10}{s(0.02s+1)(s+1)}} \cdot \frac{1}{s^2}$$

$$e_{rss} = \lim_{s \rightarrow 0} s \cdot E_r(s)$$

$$= \lim_{s \rightarrow 0} \frac{1}{s + \frac{10}{0.02s+1}(s+1)}$$

$$= 0.1$$

3. $f(t) = -u(t)$ 单独作用时 $F(s) = -\frac{1}{s}$

$$E_f(s) = -\frac{\frac{2}{s(s+1)}}{1 + \frac{10}{s(0.02s+1)(s+1)}} \cdot \left(-\frac{1}{s}\right)$$

$$e_{fss} = \lim_{s \rightarrow 0} s \cdot E_f(s)$$

$$= \lim_{s \rightarrow 0} s \cdot \frac{2}{s(s+1) + \frac{10}{0.02s+1}} \cdot \frac{1}{s}$$

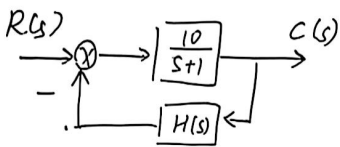
$$= 0.2$$

4. 总误差, 根据叠加原理

$$e_{ss} = e_{rss} + e_{fss} = 0.3$$



例2.



已知 $r(t) = u(t)$

求 $H(s) = 1$ 和 0.1 时的稳态误差

解: 显然稳定

$H(s) = 1$ 时

$$E(s) = \frac{1}{1 + \frac{10}{s+1}} \cdot \frac{1}{s}$$

$$e_{ss} = \lim_{s \rightarrow 0} s \cdot E(s) = \frac{1}{11} \quad E(s) = \frac{1}{11}$$

$H(s) = 0.1$ 时

$$R(s) = \frac{1}{s}$$

$$E(s) = \frac{1}{1 + \frac{10}{s+1}} \cdot \frac{1}{s} \cdot \frac{1}{0.1}$$

$$e_{ss} = \lim_{s \rightarrow 0} s \cdot E(s) = \frac{1}{0.1} = 1$$

$$E(s) = \frac{1}{H(s)}$$

二、系统的型别与参考输入的稳态误差

$$\text{开环传递函数 } G(s) = \frac{k \cdot N(s)}{s^v \cdot D(s)}$$

$$N(0) = 1, D(0) = 1$$

定义: v 型系统



用于扰动

3. $r(t) = \frac{1}{2} t^2$ 时

$$R(s) = \frac{1}{s^3} \quad SE(s) = s \cdot \frac{1}{1+G(s)} \cdot \frac{1}{s^3}$$

$\frac{E(s)}{R(s)}$ 的开环传递函数

$s=0$ 的零点的个数

$$\frac{E(s)}{R(s)} = \frac{1}{1+G(s)} = \frac{s^v D(s)}{s^v D(s) + k \cdot N(s)}$$

$$K_a = \lim_{s \rightarrow 0} s^2 G(s) = \begin{cases} 0, & v=0, 1 \\ k, & v=2 \\ \infty, & k \geq 3 \end{cases}$$

稳态加速度误差系数

1. $R(t) = u(t)$ 时 $R(s) = \frac{1}{s}, E(s) = \frac{1}{1+G(s)} R(s)$

求

$$SE(s) = s \cdot \frac{1}{1+G(s)} \cdot \frac{1}{s}$$

$$e_{ss} = \lim_{s \rightarrow 0} s E(s) = \lim_{s \rightarrow 0} \frac{1}{1+G(s)}$$

稳态位置误差系数 $K_p = \lim_{s \rightarrow 0} G(s)$

$$= \begin{cases} k, & v=0 \\ \infty, & v \geq 1 \end{cases}$$

$$e_{ss} = \begin{cases} \frac{1}{1+k_p}, & v=0 \end{cases}$$

	$Ru(t)$	Rt	$R\frac{1}{2}t^2$
0	$\frac{R}{1+k_p}$	∞	∞
I	0	$\frac{R}{k_v}$	∞
II	0	0	$\frac{R}{k_a}$



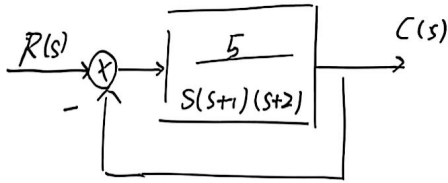
扫描全能王 创建

$e_{ss} \downarrow \rightarrow k \uparrow$ 稳定?

例5.

$e_{ss} \rightarrow 0 \rightarrow v \uparrow$

例3.



先判稳!

$r(t) = u(t), 10t, 3t^2$ 时, e_{ss} .)

$$G(s) = \frac{5}{s(s+1)(s+2)} \quad \text{一个积分环节 } \frac{1}{s}$$

~ I 型

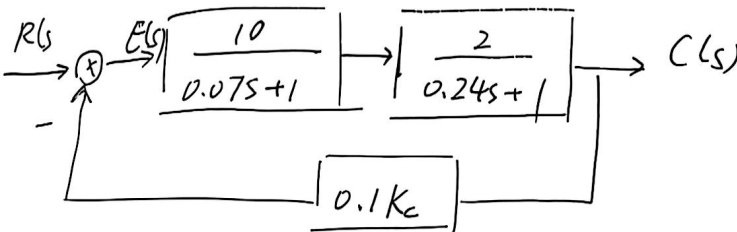
$u(t) \rightarrow e_{ss} = 0$

$3t^2 \rightarrow e_{ss} = \infty$

$10t \rightarrow k_v = \lim_{s \rightarrow 0} s G(s) = 2.5$

$$e_{ss} = \frac{10}{2.5} = 4$$

例4.



已知 $K_c = 0.05 \text{ V/(r/min)}$

求 $r(t) = u(t)$ 时 e_{ss} 误差

$$G(s) = \frac{20K_c}{(0.07s+1)(0.24s+1)} \quad \text{0 型 } v=0$$

$$K_p = \lim_{s \rightarrow 0} G(s) = 0.1$$

$$e_{ss} = \frac{1}{1 + K_p} = \frac{1}{1.1}$$

$$e_{ss} = \frac{e_{ss}}{0.1K_c} = 181.8 \text{ (r/min)}$$



定义: 频率响应

系统对正弦输入信号的稳态响应. ω 的复函数

$\omega: (-\infty, +\infty)$ 或 $[0, +\infty)$

$$\angle(\omega) = |G(j\omega)| \quad \phi(\omega) = \angle G(j\omega)$$

幅频特性

相频特性

>0 相位超前
 <0 相位滞后

例1. $G(s) = \frac{2}{s+1}$, $U_i(t) = 10 \sin t$

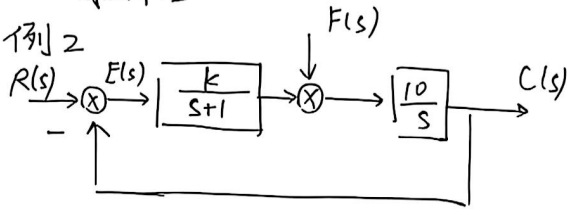
求 $U_{os}(t)$

$$G(s) = \frac{2}{s+1}, \quad G(j\omega) = \frac{2}{j\omega+1}$$

$$G(j) = \frac{2}{j+1} = \sqrt{2} e^{-j45^\circ}$$

$$U_{os}(t) = 10\sqrt{2} \sin(t-45^\circ)$$

例2



$$r(t) = t \text{ rad/s} \quad f(t) = \sin 3t \text{ rad/s}^2$$

当 $r(t)$ 单独作用时稳态误差 0.1 rad/s

求 k 和 e_{ss}

正弦信号不能用终值定理

$$V=1, \quad k_v = 10K, \quad e_{ss} = \frac{1}{10K} = 0.1 \quad K=1$$

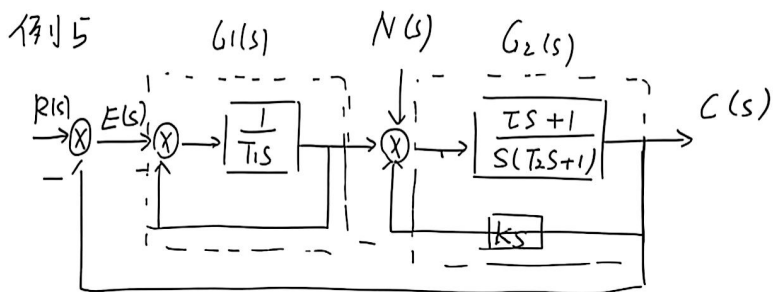
$$\omega=3, \quad \phi_{EF}(s) = \frac{s^2 + 10(s+1)}{s^2 + s + 10K} = \frac{-10(s+1)}{s^2 + s + 10}$$

$$\phi_{EF}(3j) = \frac{-10(3j+1)}{3j+1} = -10$$

$$e_{fss} = -10 \sin 3t$$

$$e_{ss} = e_{ss} + e_{fss} = -10 \sin 3t + 0.1$$





鉴别系统对 $r(t)$ 和 $n(t)$ 分别是几型系统

$$G_1(s) = \frac{T_1 s}{T_1 s + 1}$$

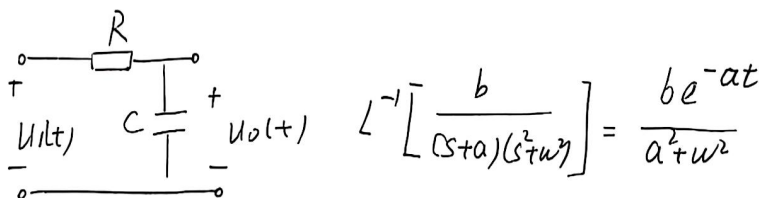
$$G_2(s) = \frac{T_2 s + 1}{T_2 s^2 + s + k T_2 s^2 + k s}$$

一个积分环节, 对 $r(t)$ I 型

$$\phi_{EN}(s) = -\frac{1}{1+G_1 G_2} \quad \text{零型}$$

第五章 线性系统的频域分析法

5.1 频率特性



$$G(s) = \frac{1}{Ts+1} + \frac{b}{\omega \sqrt{a^2 + \omega^2}} \sin(\omega t - \tan^{-1} \frac{\omega}{a})$$

$$T = RC$$

$$U_1(t) = A \cdot \sin \omega t$$

$$U_0(t) = B \cdot \sin(\omega t + \phi)$$

$$U_0(s) = \frac{1}{Ts+1} \cdot \frac{A\omega}{s^2 + \omega^2}$$

$$U_0(t) = \frac{\frac{A\omega}{T} e^{-\frac{1}{T}t}}{(\frac{1}{T})^2 + \omega^2} + \frac{\frac{A\omega}{T}}{\omega \sqrt{(\frac{1}{T})^2 + \omega^2}} \sin(\omega t - \tan^{-1} T\omega)$$

定义1:

$$G(j\omega) = G(s)|_{s=j\omega} = \frac{1}{jT\omega+1} = \frac{1}{\sqrt{1+(T\omega)^2}} e^{-j \tan^{-1} T\omega} = L(\omega) e^{j\phi(\omega)}$$

$$\begin{cases} U_0(t) = \frac{A}{\sqrt{1+(T\omega)^2}} \sin(\omega t - \tan^{-1} T\omega) \\ U_1(t) = A \cdot \sin \omega t \\ U_0(s) = A L(\omega) \sin(\omega t + \phi(\omega)) \end{cases}$$



$$G(j\omega) = G(s)|_{s=j\omega}$$

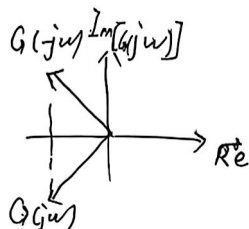
$$\omega \in (-\infty, +\infty) \text{ 或 } [0, +\infty)$$

$$|G(-j\omega)| = |G(j\omega)|$$

$$\angle G(-j\omega) = -\angle G(j\omega)$$

$$\operatorname{Re} G(j\omega) = \operatorname{Re} G(j\omega)$$

$$\operatorname{Im} G(j\omega) = -\operatorname{Im} G(j\omega)$$



定义2: $G(j\omega) = \frac{C(j\omega)}{R(j\omega)}$ 初始条件为零时

$$F[f(t)] = \int_{-\infty}^{\infty} f(t) e^{-j\omega t} dt = F(j\omega)$$

$$f(t) = \frac{a_0}{2} + \sum_{n=1}^{+\infty} [a_n \cos(n\omega t) + b_n \sin(n\omega t)]$$

$$r(t) = A \sin \omega t \quad (\omega: 0.01 \sim 100)$$

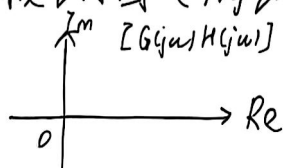
$$0.01, 0.02, \dots, 209, 0.1, 0.2, \dots, 0.9, 1, 2, \dots, 9, 10, 20, \dots, 90, 100$$

$$r(t) = \delta(t) \quad R(j\omega) = 1$$

$$c(j\omega) = G(j\omega) \cdot R(j\omega) = G(j\omega) = \int_{-\infty}^{+\infty} \underset{\substack{\uparrow \\ \text{脉冲响应}}}{k(t)} e^{-j\omega t} dt$$

5.2 典型环节的频率特性

一、极坐标图 (Nyquist 图, 幅相特性图)



1. 惯性环节

$$G(s) = \frac{1}{Ts + 1}$$

$$G(j\omega) = \frac{1}{j\omega T + 1} = \frac{1}{1 + (T\omega)^2} + j \frac{-T\omega}{1 + (T\omega)^2}$$

$$= \angle(\omega) e^{j\phi(\omega)}$$

幅频特性 相频特性

$$U(\omega) = \frac{1}{\sqrt{1 + (T\omega)^2}}$$

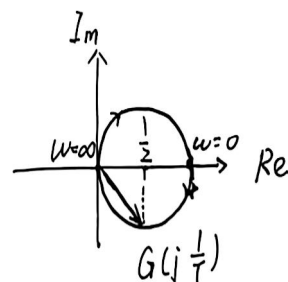
$$V(\omega) = \frac{-T\omega}{1 + (T\omega)^2}$$

$$\angle(\omega) = \frac{1}{\sqrt{1 + (T\omega)^2}}$$

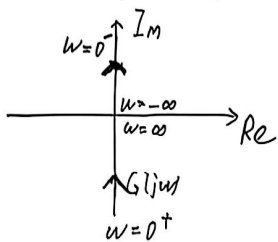
$$\phi(\omega) = -\tan^{-1} T\omega$$

$$(U - \frac{1}{2})^2 + V^2 = (\frac{1}{2})^2$$

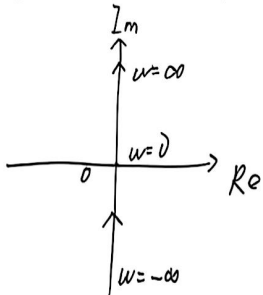
ω	$U(\omega)$	$V(\omega)$	$\angle(\omega)$	$\phi(\omega)$
0	1	0	1	0°
$\frac{1}{T}$	$\frac{\sqrt{2}}{2}$	$-\frac{\sqrt{2}}{2}$	$\frac{\sqrt{2}}{2}$	-45°
∞	0	0	0	-90°



2. 积分环节 $G(j\omega) = \frac{1}{j\omega}$

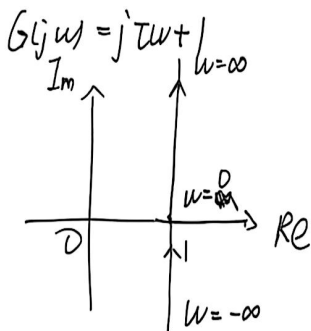


3. 纯微分环节 $G(j\omega) = j\omega$



4. 一阶微分环节

$$G(s) = Ts + 1$$



5. 振荡环节

$$G(s) = \frac{1}{T^2 s^2 + 2\zeta Ts + 1}$$

$$G(j\omega) = \frac{1}{(1 - T^2 \omega^2) + j2\zeta T\omega}$$

$$U(\omega) = \frac{1 - T^2 \omega^2}{(1 - T^2 \omega^2)^2 + (2\zeta T\omega)^2}$$

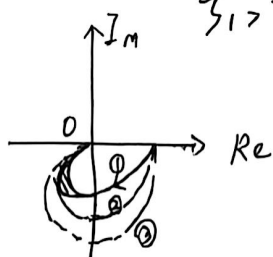
$$V(\omega) = \frac{-2\zeta T\omega}{(1 - T^2 \omega^2)^2 + (2\zeta T\omega)^2}$$

$$G(j\frac{1}{T}) = -j\frac{1}{2\zeta}$$

$$G(0) = 1$$

$$G(\infty) = 0$$

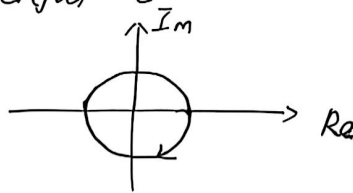
$$\angle G(j\infty) = -180^\circ$$



6. 延迟环节

$$G(s) = e^{-Ts}$$

$$G(j\omega) = e^{-jT\omega}$$



起点: $\omega \rightarrow 0^+$

$$G(s) \cdot H(s) = \frac{k(s+z_1) \cdots (s+z_m)}{s^v (s+p_{v+1}) \cdots (s+p_n)}$$

$$G(j\omega) \cdot H(j\omega) \approx \frac{k \prod_{i=1}^m z_i'}{(j\omega)^v \prod_{i=v+1}^n p_i'}$$

$$\angle G(j0^+) H(j0^+) = -90^\circ v$$

终点: $\omega \rightarrow \infty$

$$G(s) \cdot H(s) = \frac{k(s+z_1) \cdots (s+z_m)}{(s+p_1) \cdots (s+p_n)}$$

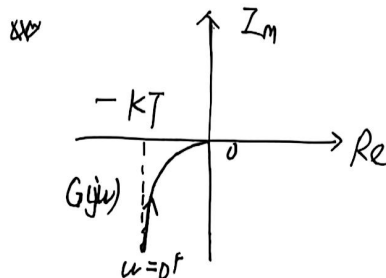
$$G(j\omega) H(j\omega) = \frac{k}{(j\omega)^{n-m}}$$

例1. $G(s) = \frac{k}{s(Ts+1)}$

例2. $G(s) = \frac{1}{(T_1 s + 1)(T_2 s + 1)(T_3 s + 1)}$

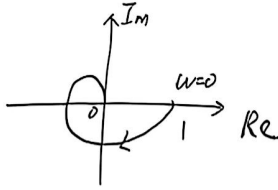
例3. $G(s) = \frac{\omega_n^2}{s(s^2 + 2\zeta \omega_n s + \omega_n^2)}$

例1. $G(j\omega) = \frac{k}{j\omega(Tj\omega+1)} = -\frac{kT}{T^2 \omega^2 + 1} - j\frac{k}{\omega(T^2 \omega^2 + 1)}$

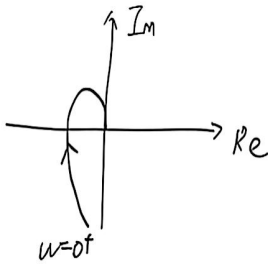


例2 $G(s) = \frac{1}{(T_1s+1)(T_2s+1)(T_3s+1)}$

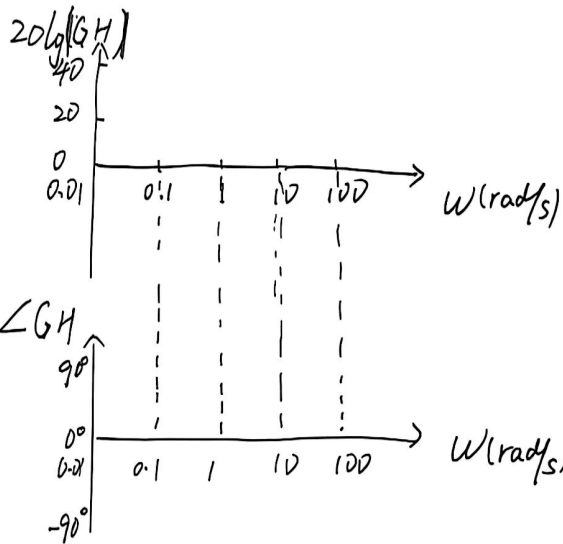
$G(j\omega) = \frac{1}{(T_1j\omega+1)(T_2j\omega+1)(T_3j\omega+1)}$



例3. $G(j\omega) = \frac{\omega n^2}{j\omega(j\omega^2 + 2\zeta n j\omega + n^2)}$



二、对数频率特性图 (Bode)



横坐标: $\lg \omega$ + 倍频程

0.01 ~ 100

$\lg 1 = 0$ $\lg 5 = 0.7$ $\lg 9 = 0.95$

$\lg 2 = 0.301$ $\lg 6 = 0.78$

$\lg 3 = 0.4771$ $\lg 7 = 0.85$

$\lg 4 = 0.602$ $\lg 8 = 0.9$

纵坐标: $20\lg |GH|$ 分贝(dB)

1. 放大环节

$G(s) = k$ $G(j\omega) = k$

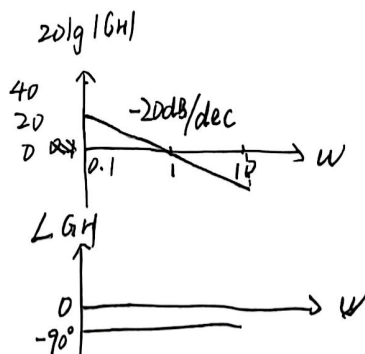
$\begin{cases} 20\lg k & \text{平行线} \times 2 \\ 0^\circ \end{cases}$

2. 积分环节

$G(s) = \frac{1}{s}$, $G(j\omega) = \frac{1}{j\omega}$

$20\lg |G(j\omega)| = -20\lg \omega$

$\angle G(j\omega) = -90^\circ$



$G(s) = \frac{1}{s^2}$, $\begin{cases} -20\text{dB/dec} \\ -180^\circ \end{cases}$



3. 惯性环节

$$G(s) = \frac{1}{Ts+1} \quad G(j\omega) = \frac{1}{jT\omega+1}$$

$$20\lg|G(j\omega)| = 20\lg \frac{1}{\sqrt{(T\omega)^2+1}}$$

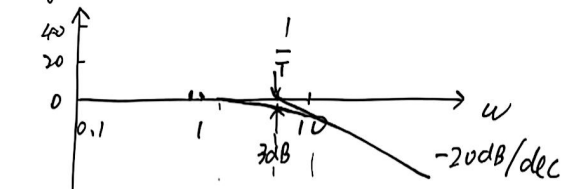
$$= \begin{cases} 0, & \omega \ll \frac{1}{T} \\ -20\lg\omega + 20\lg\frac{1}{T}, & \omega \gg \frac{1}{T} \end{cases}$$

$$\angle G(j\omega) = -\tan^{-1}T\omega$$

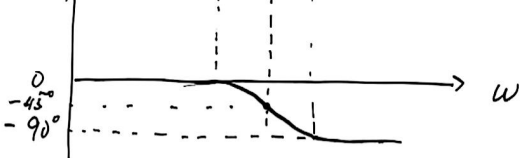
$$G(j\frac{1}{T}) = \frac{1}{j+1}$$

$$20\lg|G(j\frac{1}{T})| = 20\lg\frac{1}{\sqrt{2}} = -3dB$$

$$20\lg|G| (dB)$$



$$\angle G$$



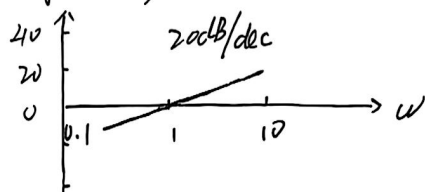
4. 纯微分环节

$$G(s) = s \quad G(j\omega) = j\omega$$

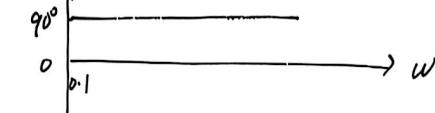
$$20\lg|G(j\omega)| = 20\lg\omega$$

$$\angle G(j\omega) = 90^\circ$$

$$20\lg|G| (dB)$$



$$\angle G$$



5. 二阶微分环节

$$G(s) = Ts+1 \quad G(j\omega) = jT\omega+1$$

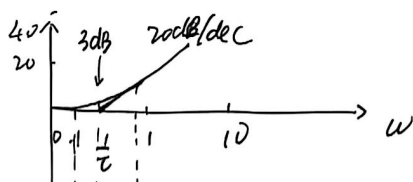
$$20\lg|G(j\omega)| = 20\lg\sqrt{(T\omega)^2+1} = \begin{cases} 0, & \omega \ll \frac{1}{T} \\ 20\lg\omega + 20\lg\frac{1}{T}, & \omega \gg \frac{1}{T} \end{cases}$$

$$G(j\frac{1}{T}) = \frac{1}{j+1}$$

$$20\lg|G(j\frac{1}{T})| = 20\lg\frac{1}{\sqrt{2}} = -3dB$$

$$20\lg|G(j\frac{1}{T})| = -3dB$$

$$20\lg|G(j\frac{1}{T})| = -3dB$$



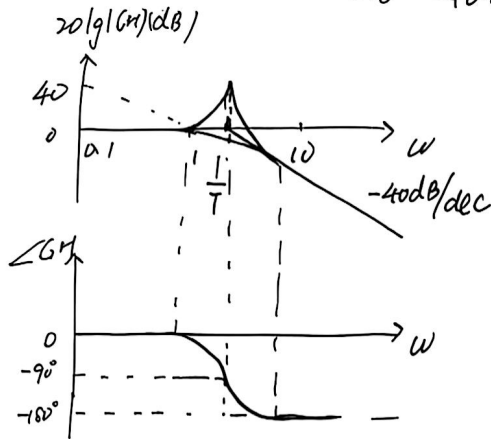
6. 振荡环节

$$G(s) = \frac{1}{T^2 s^2 + 2\zeta Ts + 1} \quad G(j\omega) = \frac{1}{1 - T^2 \omega^2 + j2\zeta T\omega}$$

$$20\lg|G(j\omega)| = 20\lg \frac{1}{\sqrt{(1 - T^2 \omega^2)^2 + (2\zeta T\omega)^2}} = \begin{cases} 0, & \omega < \frac{1}{T} \\ -40\lg\omega + 40\lg\frac{1}{T}, & \omega > \frac{1}{T} \end{cases}$$

$$G(j\frac{1}{T}) = \frac{1}{j2\zeta} \quad \begin{matrix} \zeta = 0.2 & 8\text{dB} \\ 0.5 & 0\text{dB} \\ 0.8 & -4\text{dB} \end{matrix}$$

$$\angle G(j\omega) = -\arctan \frac{2\zeta T\omega}{1 - T^2 \omega^2}$$



7. 二阶微分环节

与振荡相反

$$G(s) = T^2 s^2 + 2\zeta Ts + 1$$

延迟环节
8. $G(s) = e^{-Ts}$

$$G(j\omega) = e^{-jT\omega}$$

$$20\lg|G(j\omega)| = 20\lg 1 = 0\text{dB}$$



$$G(s) = G_1(s) G_2(s) \dots G_n(s)$$

$$20\lg|G(j\omega)| = 20\lg|G_1 G_2 \dots G_n|$$

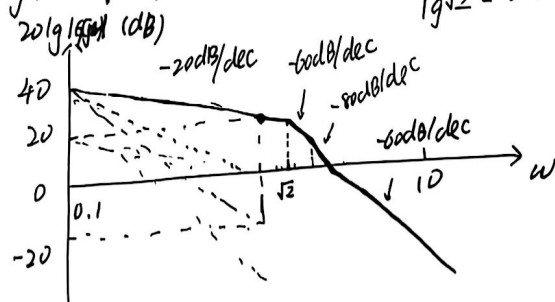
$$= 20\lg[|G_1(j\omega)| \cdot |G_2(j\omega)| \cdot \dots \cdot |G_n(j\omega)|]$$

$$= 20\lg|G_1(j\omega)| + \dots + 20\lg|G_n(j\omega)|$$

例1. $G(s) = \frac{10(\frac{1}{3}s + 1)}{s(\frac{1}{2}s + 1)(\frac{1}{2}s^2 + s + 1)}$

$$20\lg 10 = 20\text{dB}$$

转折频率: 3, 2, $\sqrt{2}$ rad/s



$$\frac{s^2 + 2s + 1}{(s+1)^2} = 1$$

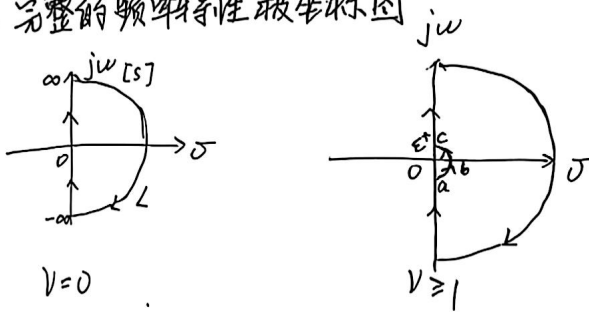
$$\frac{0.15 \times 0.9}{0.1 \times 1.0} = 1.35$$

$$\lg \sqrt{2} = 0.15 \quad \lg 3 = 0.4771$$

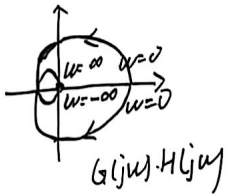


5.3 Nyquist 稳定判据

一. 完整的频率特性极坐标图



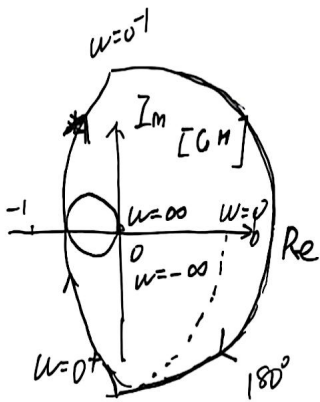
1. $V=0$



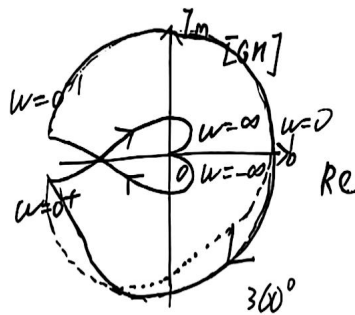
2. $V \geq 1$

$\widehat{abc} \quad s = \xi^+ e^{j\theta}$
 $\xi^+ \rightarrow 0^+$
 $\theta: -90^\circ \rightarrow 0^\circ \rightarrow 90^\circ$
 $G(s)H(s) \approx \frac{K}{s^V} = \frac{k}{(\xi^+)^V} e^{-j\theta V}$

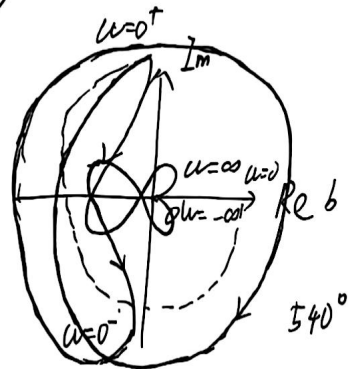
	s	$\theta(\text{rad})$	$ GH $	$\angle GH$
a	$j0^-$	$-\frac{\pi}{2}$	∞	$\frac{\pi}{2}V$
b	0	0	∞	0
c	$j0^+$	$\frac{\pi}{2}$	∞	$-\frac{\pi}{2}V$



$V=1$



$V=2$



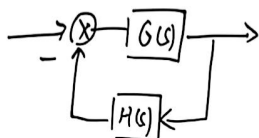
$V=3$

虚线 $0 \rightarrow +\infty$

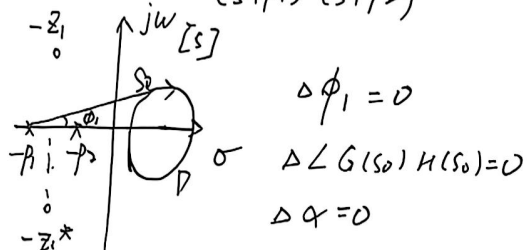
实线 + 圆 $0^- \rightarrow +\infty$



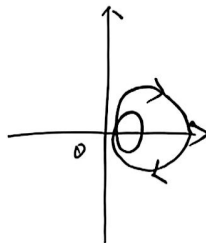
二、幅角定理



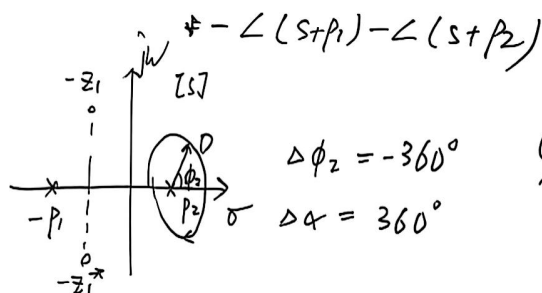
$$G(s) \cdot H(s) = \frac{k(s+z_1)(s+z_1^*)}{(s+p_1)(s+p_2)}$$



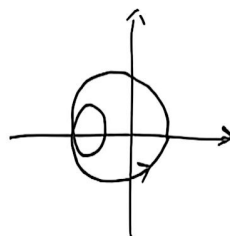
G_H



$$\alpha = \angle G(s_0)H(s_0) = \angle(s+z_1) + \angle(s+z_1^*)$$



G_H



若某复变函数在封闭曲线 D 的内部有 z 个零点和 p 个极点，则 D 经该复变函数映射后的曲线包围原点 $z-p$ 圈（ D ，映射后同顺同逆 负号反向）

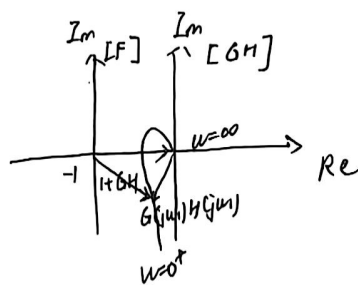
三、Nyquist 稳定判据

$$G(s)H(s) = \frac{b_ms^m + \dots + b_1s + b_0}{s^v(s-p_{v+1})\dots(s-p_n)}$$

$$F(s) = 1 + G(s)H(s) = \frac{k(s-\hat{z}_1)\dots(s-\hat{z}_n)}{s^v(s-p_{v+1})\dots(s-p_n)}$$

p_i — 开环极点

$\hat{z}_i$ — 闭环极点



$z=0$
对应开环极点
闭环极点 $z-p$ ($F(s)$ 的)

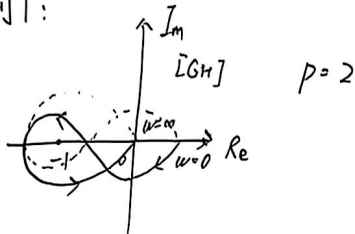
[F]: 闭环系统稳定的充要条件：当 s 沿 Nyquist 曲线顺时针连续运动一周时， D 在 $[F]$ 的映射 逆时针 包围原点 P 圈

[G] 奈氏稳定判据(I) 闭环系统稳定的充要条件：当 ω 由 $-\infty \rightarrow \infty$ 变化时，开环频率特性 $G(j\omega)H(j\omega)$ 的奈氏图 逆时针 包围 $(-1, j0)$ 点 P 圈

(II) $0 \rightarrow \infty$ $\frac{P}{2}$ 圈



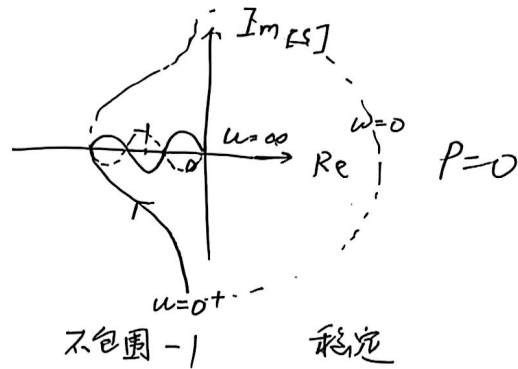
例1:



绕-1逆时针2圈

稳定

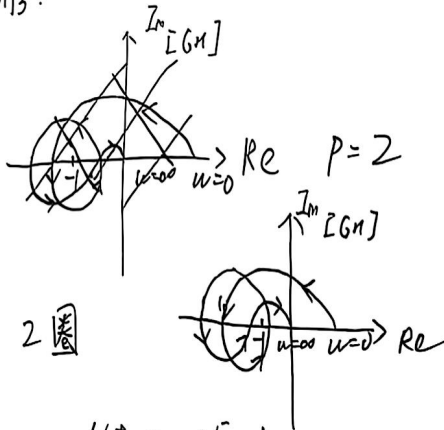
例2:



不包围-1

稳定

例3:

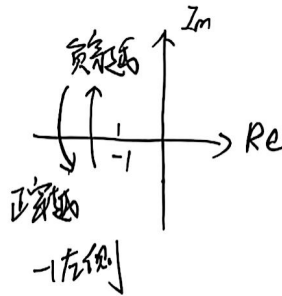


2圈

$$N^+ = 2, N^- = 1$$

$$N^+ - N^- = \frac{P-Z}{2}$$

$$2-1 = \frac{1}{2}(2-Z) \quad Z=0$$



(I) $\omega: 0 \rightarrow \infty$ 时

正负穿越次数之差为 $\frac{P}{2}$

$$N^+ - N^- = \frac{P-Z}{2} \quad (\text{稳定})$$

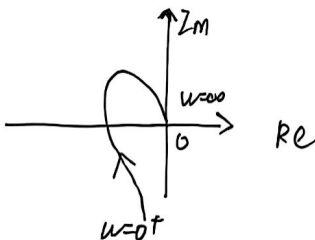
不稳定闭环
极点数为

例4: $G(s)H(s) = \frac{k}{s(T_1s+1)(T_2s+1)}$

$$K>0, T_1>0, T_2>0$$

$$\rightarrow P=0$$

正实根个数=P

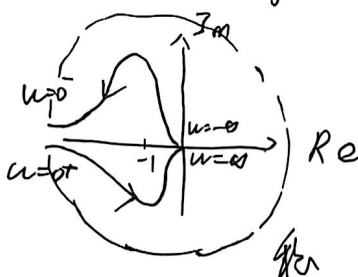


顺时针1圈

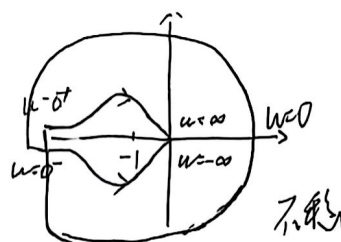
例5: $G(s)H(s) = \frac{k(Ts+1)}{s^2(Ts+1)} \quad P=0$

$$\angle G(j\omega)H(j\omega) = -180^\circ + \tan^{-1}\tau\omega - \tan^{-1}T\omega$$

$$\tau > T$$



稳定

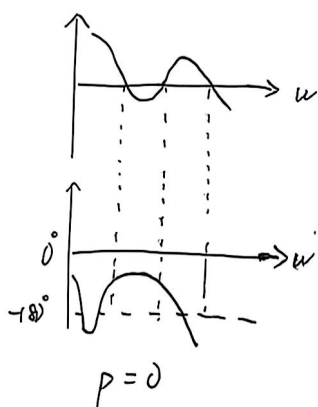
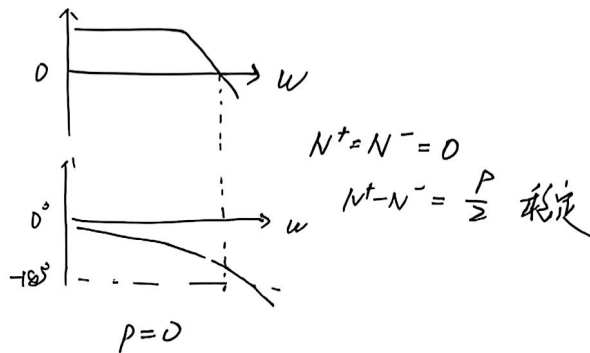
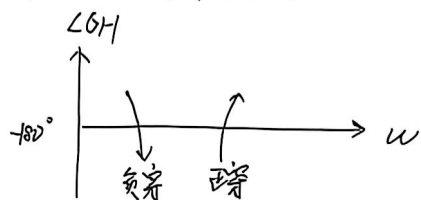


不稳定



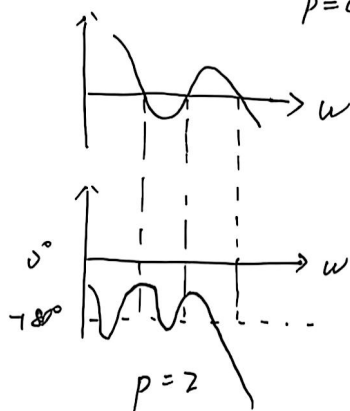
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四、用Bode图判断闭环系统稳定性



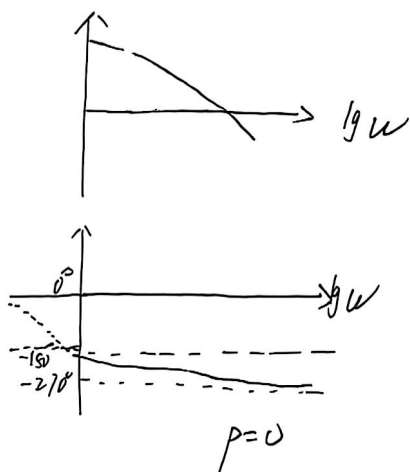
$$N^+ = 1, N^- = 2$$

不稳 $Z=2$



$$N^+ = 1, N^- = 2$$

不稳 $Z=4$



2个积分环节
 $\omega \rightarrow 0^+ \angle GH \rightarrow 180^\circ$
 $\omega \rightarrow \infty \angle GH \rightarrow 0^\circ$
 $N^- = 1, N^+ = 0$

$Z=2$

5.4 控制系统的相对稳定性

一、相位裕度

1. 剪切频率 (ω_c)

$$|G(j\omega_c)H(j\omega_c)| = 1$$

2. 相位裕度

$$\gamma = \angle G(j\omega_c)H(j\omega_c) - (-180^\circ)$$

$$\therefore \gamma = 180^\circ + \angle G(j\omega_c)H(j\omega_c)$$

二、幅值裕度

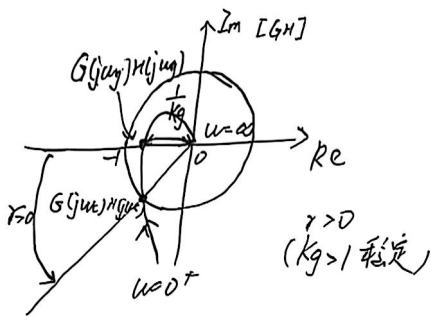
1. 穿越频率 (ω_g)

$$\angle G(j\omega_g)H(j\omega_g) = -180^\circ$$

2. 幅值裕度

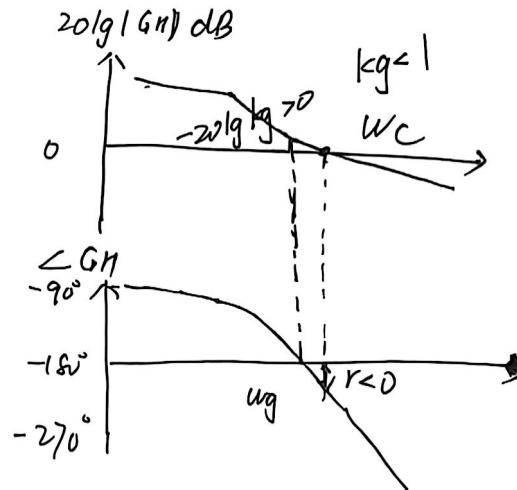
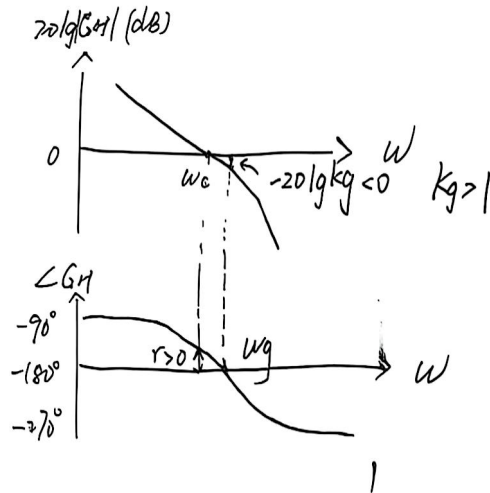
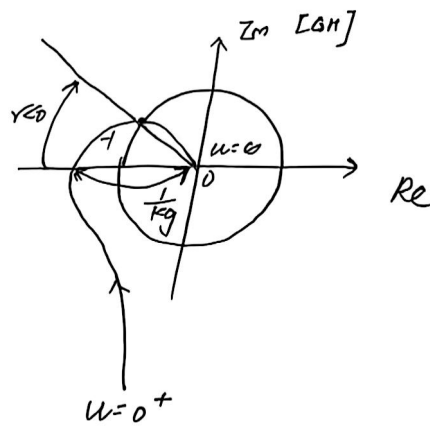
$$K_g = \frac{1}{|G(j\omega_g)H(j\omega_g)|}$$



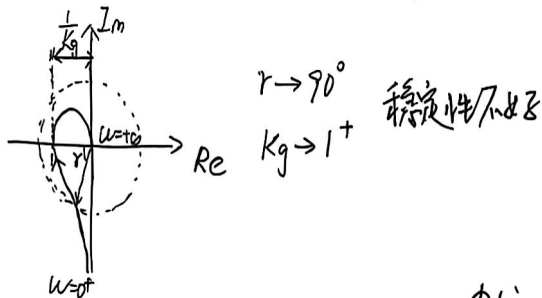


$$\gamma = 40^\circ \sim 60^\circ$$

$$K_g = 2 \sim 3.16 = 6 \sim 10 \text{ dB}$$

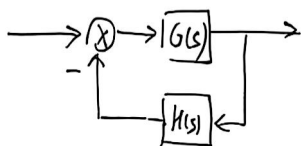


例 1. $G(s)H(s) = \frac{1}{S(T^2s^2 + 2\zeta Ts + 1)}$



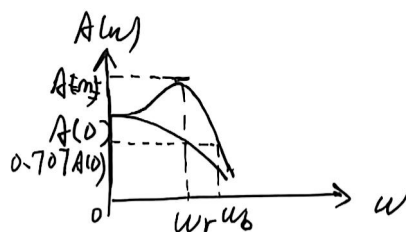
5.5 闭环频率特性

$$\phi(j\omega) = \begin{cases} \frac{1}{H(j\omega)}, & |G(j\omega)H(j\omega)| \gg 1 \text{ 低频} \\ G(j\omega), & |G(j\omega)H(j\omega)| \ll 1 \text{ 高频} \end{cases}$$



$$\phi(s) = \frac{G(s)}{1 + G(s)H(s)}$$

$$\begin{aligned} \phi(j\omega) &= \frac{G(j\omega)}{1 + G(j\omega)H(j\omega)} \\ &= A(\omega) e^{j\theta(\omega)} \end{aligned}$$



ω_r — 谐振频率

$$M_r = \frac{A_{\text{max}}}{A(0)} \text{ 谐振峰值}$$

ω_b 截止频率 (带宽)

$A(\omega)$ 闭环频率特性的幅频特性

$\theta(\omega)$ 闭环频率特性的相频特性



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5.6 开环频率特性与控制系统性能的关系

一、控制系统的性能指标

1. 时间域指标

(1) 动态指标: $\begin{matrix} t_r & t_p & t_s \\ \sigma & N & \end{matrix}$

(2) 稳态指标: e_{ss}, ν, K

2. 频率域指标

(1) 开环指标: $\gamma, K_g, \omega_c, \omega_g$

ω_b, ω_c 同数量级

(2) 闭环指标: ω_r, M_r, ω_b

二、经验公式

$$1. M_r = \frac{1}{\sin \gamma}$$

$$2. \sigma_p = 0.16 + 0.4(M_r - 1)$$

$$(1 \leq M_r \leq 1.8)$$

$$3. t_s = \frac{\pi}{\omega_c} [2 + 1.5(M_r - 1) + 2.5(M_r - 1)^2] \quad \text{带宽越宽, 响应速度越快}$$

$$(0 < M_r \leq 1.8)$$

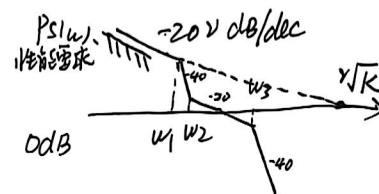
三、开环对数幅频特性与性能指标的关系

1. 低频段 $\omega < \omega_1$

$$G(s)H(s) = \frac{K}{s^\nu}$$

$$G(j\omega)H(j\omega) = \frac{K}{(j\omega)^\nu}$$

$$20 \lg |G(j\omega)H(j\omega)| = 20 \lg K - 20\nu \lg \omega = 0 \quad \omega = \sqrt[\nu]{K}$$



2. 中频段 $\omega_2 < \omega < \omega_3$

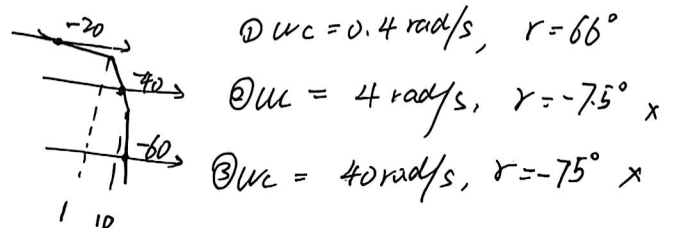
$$t_s = \frac{\pi}{\omega_c} * [2 + 1.5(M_r - 1) + 2.5(M_r - 1)^2]$$

例: $G(s)H(s) = \frac{K}{s(s+1)(0.1s+1)}$

$$\gamma = 180^\circ + \angle G(j\omega_c)H(j\omega_c)$$

$$= 180^\circ - 90^\circ - \tan^{-1} \omega_c - \tan^{-1} 0.1 \omega_c$$

$$= 90^\circ - \tan^{-1} \omega_c - \tan^{-1} 0.1 \omega_c$$



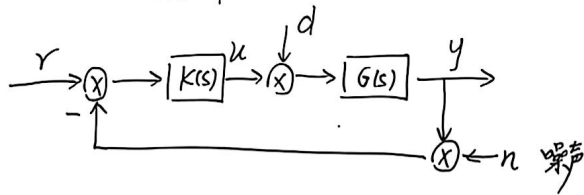
以 -20 dB/dec 的斜率过 0 dB 线

$$\omega_2 \leq \omega_c \frac{Mr-1}{Mr} \quad \omega_3 \geq \omega_c \frac{Mr+1}{Mr} \quad \lambda = \frac{\omega_3}{\omega_2} \geq \frac{Mr+1}{Mr-1}$$



5.7 控制系统设计初步概念 (不考)

一、控制系统的矛盾因素与限制



$$y = \frac{KG}{1+KG}(r-n) + \frac{G}{1+KG}d$$

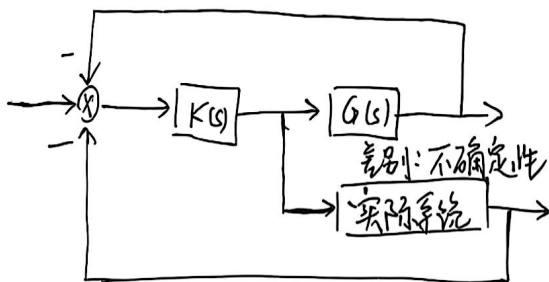
$$\text{误差 } e \triangleq r-y = \frac{1}{1+KG}r - \frac{G}{1+KG}d + \frac{KG}{1+KG}n$$

$$|K(j\omega)G(j\omega)| \gg 1$$

$$e \approx \frac{1}{1+KG}r - \frac{G}{1+KG}d + n$$

限制高增益

- ① 噪声误差
- ② 过量的 u
- ③ 要允许不确定性



$$u = \frac{k}{1+KG}(r-n) - \frac{KG}{1+KG}d$$

低频 $|KG| \gg 1$

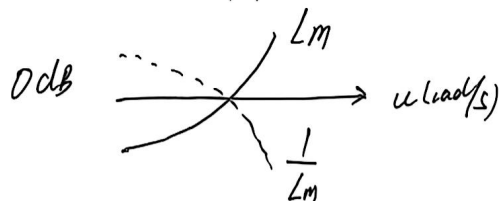
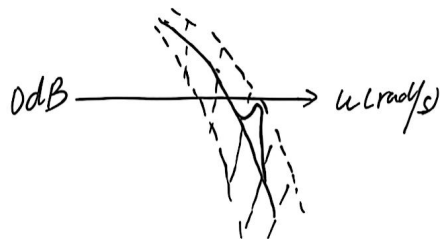
$$u \approx \frac{1}{G}(r-n) - \frac{KG}{1+KG}d$$

$$u \uparrow \rightarrow G \downarrow \rightarrow u \uparrow$$

1. 乘性不确定性

$$G(j\omega) = [1 + L(j\omega)]G_0(j\omega)$$

$$|L(j\omega)| < L_m(\omega)$$



二、对象的不确定性

1. 造成不确定性的因素

(1) 参数变化:

(2) 未建模动态特性

① 谐振 ② 延迟 ③ 分布参数

2. 不确定性的描述

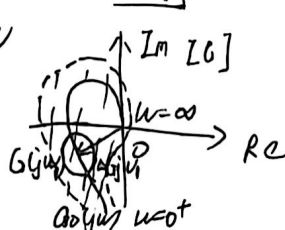
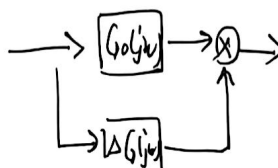
(1) 加性不确定性

$$G(j\omega) = G_0(j\omega) + \Delta G(j\omega)$$

实际特性 名义特性 不确定性

$$|\Delta G(j\omega)| < L_a(\omega)$$

界函数

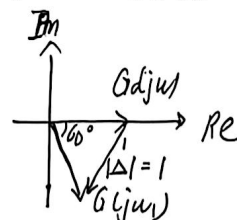


例: 未建模 e^{-Ts}

$$G_0(s) = 1 \quad T = 0.05 \text{ sec} \quad \omega_1 = 20 \text{ rad/sec}$$

$$G(j\omega) = e^{-j\omega T}$$

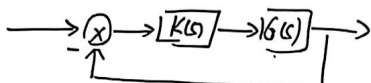
$$\varphi(\omega_1) = -0.05 \times 20 = -1 \text{ rad} \approx -60^\circ$$



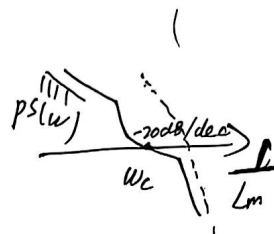
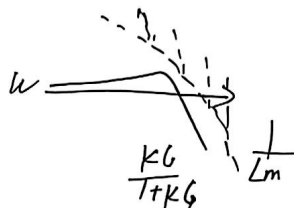
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三、设计准则

鲁棒稳定性：在扰动情况下仍保证稳定



鲁棒稳定性条件 $\left| \frac{KG}{1+KG} \right| < \frac{1}{L_m}$

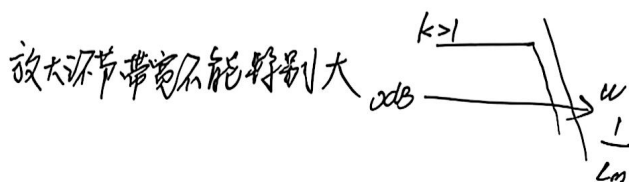


1. 名义系统稳定

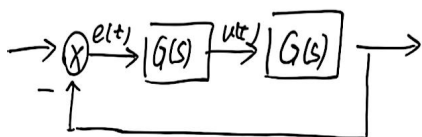
2. 低频段增益要高于跟踪误差和干扰抑制所要求性能指标 $P_s(\omega)$

3. 高频段增益要低于不确定性和噪声所规定的界 $\frac{1}{L_m}$

4. 以 -20dB/dec 的斜率过 0dB 线



5.8 PID 控制器



$$G(s) = K_p + \frac{K_i}{s} + K_d s$$

$$= K_p \left(1 + \frac{1}{T_i s} + T_d s \right)$$

一、比例控制规律 (P)

$$u(t) = K_p \cdot e(t)$$

$$G(s) = \frac{1}{Ts}$$

$$K_p = 1 \quad \phi(s) = \frac{1}{Ts+1}$$

$$K_p > 1 \quad \phi(s) = \frac{\frac{K_p}{Ts+1}}{1 + \frac{K_p}{Ts+1}} = \frac{1}{\frac{Ts}{K_p} + 1}$$

$\frac{T}{K_p} < T$ 闭环极点 $-\frac{K_p}{T} < -\frac{1}{T}$ 离虚轴越远 响应速度越快

① $T \downarrow \rightarrow t_s \downarrow$

② $K_p \uparrow \rightarrow e_{ss} \downarrow \rightarrow \text{精度} \uparrow \rightarrow \text{相对稳定性} \downarrow$



二、比例加微分(PD)

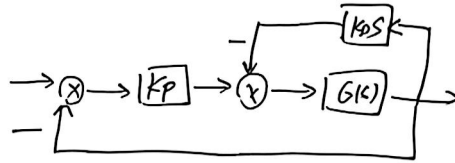
$$u(t) = K_p e(t) + K_D \frac{de(t)}{dt}$$

$$G(s) = K_p + K_D s$$

① 预见性 (D)

② $G_c(j\omega) > 0$ 相角裕度 $\gamma \uparrow$ 有助于改变动态特性

直接 G(s) PD 或 $\downarrow$ 有突变时下面影响小



微分项对噪声有放大作用 (缺点)

例 1. $G(s) = \frac{1}{Js^2}$

$$\phi(s) = \frac{\frac{1}{Js^2}}{1 + \frac{1}{Js^2}} = \frac{1}{Js^2 + 1}$$

$$D(s) = Js^2 + 1 = 0$$

$$s_{1,2} = \pm j\sqrt{\frac{1}{J}}$$

$\xi = 0$ 不正常工作
振荡

$$\phi'(s) = \frac{K_p + K_D s}{Js^2} = \frac{K_D s + K_p}{Js^2 + K_D s + K_p}$$

$$D'(s) = s^2 + \frac{K_D}{J}s + \frac{K_p}{J}$$

$$= s^2 + 2\xi\omega_n s + \omega_n^2$$

$\xi \neq 0$

③ 提高 $\xi \uparrow$

三、积分 (I)

$$u(t) = K_i \int e(t) dt$$

$$G(s) = \frac{K_i}{s}$$

无差度 $\uparrow$ 型别 $\uparrow$

① 减小甚至消除 e.s.s., 提高稳态性能

② $\angle G_c(j\omega) = -90^\circ$ $\gamma \downarrow$ 相对稳定性降低

四、PI 控制

$$u(t) = K_p e(t) + K_i \int e(t) dt$$

$$G(s) = K_p + \frac{K_i}{s} = \frac{K_p s + K_i}{s}$$

相位裕度减少的小一些

五、PID 控制

$$u(t) = K_p e(t) + K_i \int e(t) dt + K_D \frac{de(t)}{dt}$$

$$G(s) = K_p + \frac{K_i}{s} + K_D s$$

$$= \frac{K_D s^2 + K_p s + K_i}{s}$$

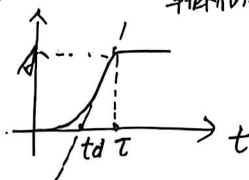
$\angle G_c(j\omega) > 0$ 型别 $\uparrow$



调整PID参数方法

Ziegler 和 Nichols (Z-N法)

$$G(s) = \frac{Ae^{-tds}}{\tau s + 1}$$



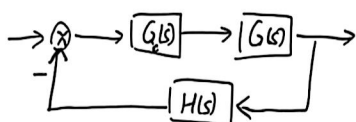
极限灵敏度法

临界稳定时, 增益 k_u (极限增益)

振荡周期 P_u (极限周期)

$$\begin{cases} P: k_p = 0.5 k_u \\ PI: k_p = 0.45 k_u, T_i = \frac{P_u}{1.2} \\ PID: k_p = 0.6 k_u, T_i = \frac{P_u}{2}, T_d = \frac{1}{8} P_u \end{cases}$$

5.9 超前补偿



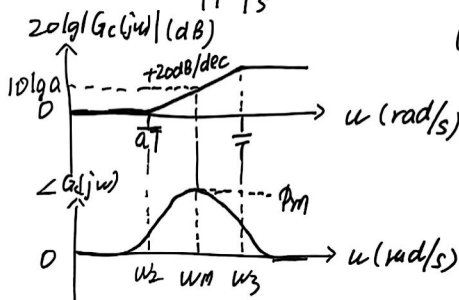
$$\begin{aligned} \tan \angle G_c(j\omega) &= \frac{\tau\omega - T\omega}{1 + \tau T\omega^2} \\ &= \frac{aT\omega - T\omega}{1 + aT^2\omega^2} \end{aligned}$$

$$G_c(s) = \frac{1 + \tau s}{1 + Ts} \quad (\tau > T)$$

$$\angle G_c(j\omega) = \tan^{-1} \tau\omega - \tan^{-1} T\omega > 0$$

$$\tau = aT \text{ 时}$$

$$G_c(s) = \frac{1 + aTs}{1 + Ts} \quad (a > 1)$$



$$\begin{aligned} \phi(\omega) &= \angle G_c(j\omega) \\ &= \arctan \frac{aT\omega - T\omega}{1 + aT^2\omega^2} \\ \omega_m &= \frac{1}{\sqrt{aT}} = \sqrt{\omega_2 \omega_3} \end{aligned}$$

$$\begin{aligned} \phi_m &= \arctan \frac{a-1}{2\sqrt{a}} \\ &= \arcsin \frac{a-1}{a+1} \\ a &= \frac{1 + \sin \phi_m}{1 - \sin \phi_m} \end{aligned}$$

例1. $G_0(s) = \frac{k}{s(0.001s+1)(0.1s+1)}$

要求 (1) 响应的速度信号 $r(t) = Rt$ 时稳态误差

不大于 $0.001R$, R 为常量

(2) $\omega_c = 165 \text{ rad/s}$

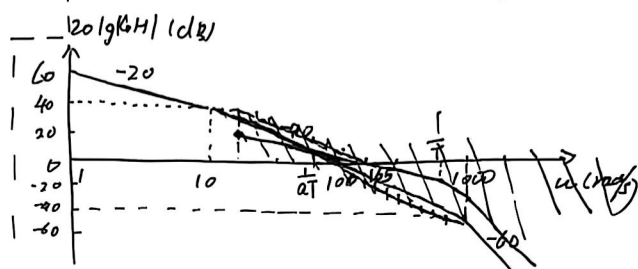
(3) $\gamma = 45^\circ$

(4) $20 \lg k_g \geq 15 \text{ dB}$

解: ① $V=1$ $e_{ss} = \frac{R}{k} \leq 0.001R$

$\therefore k \geq 1000$ 取 $k=1000$

$$G_0(s) = \frac{1000}{s(0.001s+1)(0.1s+1)}$$



校正 $\omega_{c0} = 100 \text{ rad/s}$ $\gamma = 0^\circ$

$$\textcircled{2} G_c(s) = \frac{1 + aTs}{1 + Ts} \quad (a > 1)$$

③ 取 $\omega_m = \omega_c = 165 \text{ rad/s}$

$$\therefore \gamma = 180^\circ + \angle G_0(j\omega_c) + \angle G_c(j\omega_c)$$

$$\phi_m = \angle G_c(j\omega_c) = \gamma - (180^\circ + \angle G_0(j\omega_c)) + \epsilon$$

(在 ω_c 处 -40 dB/dec , $\epsilon = 5 \sim 10$, -60 dB/dec , $\epsilon = 15 \sim 20$)

$$= 51^\circ + 5^\circ = 56^\circ$$

$$a = \frac{1 + \sin \phi_m}{1 - \sin \phi_m} = 10.7 \text{ 取 } 10$$

④ 方法一. $\omega_m = \frac{1}{\sqrt{aT}} = 165 \text{ rad/s}$

$$T = 0.00192 \text{ sec}$$

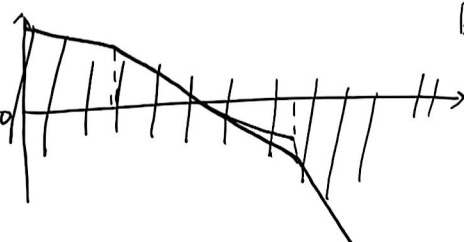
$$G_c(s) = \frac{1 + 0.0192s}{1 + 0.00192s}$$

⑤ 检验

$$G(s) = G_0(s) G_c(s) = \dots$$

$$\gamma = 47.4^\circ \quad K_g = 16.8 \text{ dB}$$

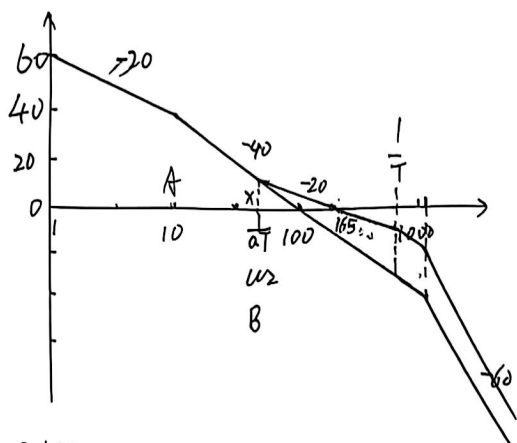
经检验, 满足要求



图见背面



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④方法二

$$\begin{cases} \frac{x-0}{\lg \omega_2 - \lg 100} = -40 \\ \frac{x-0}{\lg \omega_2 - \lg 165} = -20 \end{cases} \quad aT = \frac{1}{\omega_2} = 0.0165$$

$$G(s) = \frac{1+0.0165s}{1+0.00165s}$$

④方法三

量得 AB 长 23.6cm
+ 低频程为 30cm
 $10 \frac{23.6}{30} = 6.12$
 $\frac{1}{aT} = 6.12$

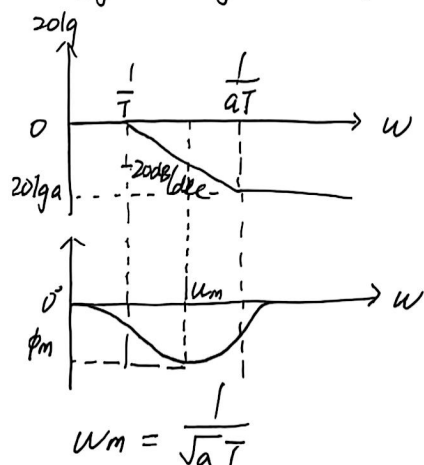
$$aT = 0.016$$

$$G(s) = \frac{1+0.016s}{1+0.0016s}$$

5.10 滞后校正

$$G_c(s) = \frac{1+aTs}{1+Ts} \quad (a < 1)$$

$$\angle G_c(j\omega) = \lg^{-1} aT\omega - \lg^{-1} T\omega < 0$$



$$\omega_m = \frac{1}{\sqrt{aT}}$$

$$\phi_m = -\arcsin \frac{1-q}{1+q}$$

$$\frac{1}{aT} = (0.1 \sim 0.2) \omega_c$$

③方法二

$$\begin{cases} \frac{20\lg 25 - y}{\lg \omega_1 - \lg \omega_2} = -20 \\ \frac{y - x'}{\lg \omega_1 - \lg \omega_2} = -40 \end{cases} \quad \frac{0-x'}{\lg 25 - \lg 0.25} = -20$$

$$\therefore x' = 20 \lg \frac{25}{0.25} = 20$$

$$\text{例 1. } G_0(s) = \frac{Kv}{s(0.1s+1)(0.2s+1)}$$

要求: (1) $v=1$

$$(2) Kv = 25 \text{ sec}^{-1}$$

$$(3) \omega_c = 2.5 \text{ rad/s}$$

$$(4) \gamma \geq 40^\circ$$

$$\text{解: } ① G_0(s) = \frac{25}{s(0.1s+1)(0.2s+1)}$$

$$20\lg 25 = 28$$

$$\angle G_0(j\omega) = -90^\circ - \lg^{-1} 0.1\omega - \lg^{-1} 0.2\omega$$

$$\gamma = 180^\circ + \angle G_0(j\omega_c)$$

$$= 49.4^\circ > 40^\circ$$

$$② G_c(s) = \frac{1+aTs}{1+Ts} \quad (a < 1)$$

$$③ \frac{20\lg 25 - x}{\lg 1 - \lg 25} = -20 \quad x = 20\lg 25 - 20\lg 2.5 = 20\text{dB}$$

下移 20dB

$$20\lg a = -20 \quad a = 0.1$$

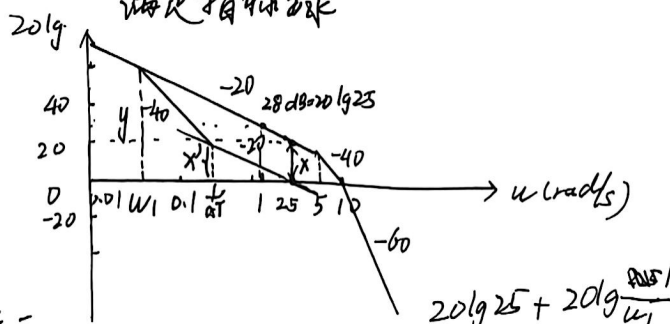
$$④ \text{取 } \frac{1}{aT} = 0.1 \omega_c \quad T = 40\text{sec}$$

$$⑤ G_c(s) = \frac{1+4s}{1+40s}$$

⑥ 检验

$$\omega_c = 2.23 \text{ rad/s} \quad \gamma = 53.8^\circ$$

满足指标要求



$$20\lg 25 + 20\lg \frac{0.1}{\omega_1}$$

$$= 20 - 40\lg \frac{\omega_1}{0.25}$$

$$\therefore \omega_1 = 0.025 \quad T = 40\text{sec}$$

⑦方法三 量



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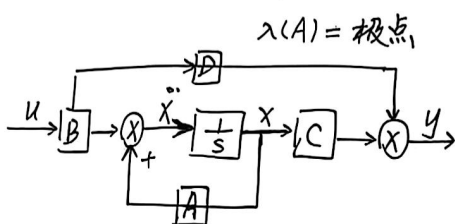
第八章 现代控制理论基础

8.1 状态空间法的基本概念

状态变量：独立变量，数目最小

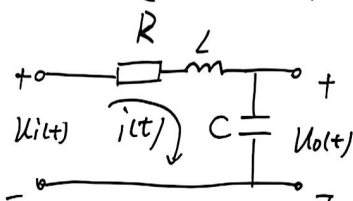
状态向量：状态空间表达式：

$$x(t) = \begin{bmatrix} x_1(t) \\ \vdots \\ x_n(t) \end{bmatrix} \quad \begin{cases} \dot{x} = Ax + Bu & \text{状态方程} \\ y = Cx + Du & \text{输出方程} \end{cases}$$



$\lambda(A) = \text{极点}$

8.2 线性定常系统状态空间表达式的建立



$$\begin{cases} Ri(t) + L \frac{di(t)}{dt} + u_o(t) = u_i(t) \\ i(t) = C \frac{du_o(t)}{dt} \end{cases} \quad u = u_i, y = u_o$$

取 $x_1(t) = u_o(t)$ $x_2(t) = i(t)$

$$\begin{cases} \dot{x}_1(t) = \frac{1}{C} x_2(t) \\ \dot{x}_2(t) = -\frac{1}{L} x_1 - \frac{R}{L} x_2 + \frac{1}{L} u_i(t) \end{cases}$$

令 $x = \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix}$

$$\begin{cases} \dot{x}(t) = \begin{bmatrix} 0 & \frac{1}{C} \\ -\frac{1}{L} & -\frac{R}{L} \end{bmatrix} x(t) + \begin{bmatrix} 0 \\ \frac{1}{L} \end{bmatrix} u(t) \\ y(t) = [1, 0] x(t) \end{cases}$$

二、根据微分方程和传递函数建立状态空间表达式

1. 传递函数没有零点 (不含输入信号导数项)

$$y^{(n)}(t) + a_1 y^{(n-1)}(t) + \dots + a_n y(t) = b_0 u(t)$$

$$G(s) = \frac{b_0}{s^n + a_1 s^{n-1} + \dots + a_n}$$

取 $x_1(t) = y(t)$

$\dots x_n(t) = \dot{x}_{n-1}(t) = y^{(n-1)}(t)$

$x_2(t) = \dot{x}_1(t) = \dot{y}(t)$

$x_3(t) = \dot{x}_2(t) = \ddot{y}(t)$

$\dot{x}_1(t) = x_2(t) \quad \dot{x}_2(t) = x_3(t)$

$\dots \dot{x}_{n-1}(t) = x_n(t)$

$\dot{x}_n(t) = y^{(n)}(t) = -a_n x_1 - a_{n-1} x_2 - \dots - a_1 x_n + b_0 u$

令 $x(t) = \begin{bmatrix} x_1(t) \\ x_2(t) \\ \vdots \\ x_n(t) \end{bmatrix}$ 可观测规范型

$$\begin{cases} \dot{x}(t) = \begin{bmatrix} 0 & 1 & 0 & \dots & 0 \\ 0 & 0 & 1 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & 1 \\ -a_n & -a_{n-1} & \dots & -a_2 & -a_1 \end{bmatrix} x(t) + \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ b_0 \end{bmatrix} u(t) \\ y(t) = [1 \ 0 \ 0 \ \dots \ 0] x(t) \end{cases}$$

例 8.1 $\ddot{y} + 5\dot{y} + 2y = 2u$

取 $x_1(t) = y(t)$

$\dot{x}_1(t) = x_2(t)$

$x_2(t) = \dot{x}_1(t) = \dot{y}(t) \quad \dot{x}_2(t) = x_3(t)$

$x_3(t) = \dot{x}_2(t) = \ddot{y}(t) \quad \dot{x}_3(t) = \ddot{y} = -5\dot{y} - 2y + 2u$

$= -5x_3 - 2x_1 + 2u$

令 $x(t) = \begin{bmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \end{bmatrix}$

$$\begin{cases} \dot{x}(t) = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -2 & 0 & -5 \end{bmatrix} x(t) + \begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix} u(t) \\ y(t) = [1 \ 0 \ 0] x(t) \end{cases}$$

2. 传递函数有零点

$$\begin{aligned} & y^{(n)}(t) + a_1 y^{(n-1)}(t) + \dots + a_n y(t) \\ & = b_0 u^{(n)}(t) + \dots + b_n u(t) \end{aligned}$$

方法一： $x_1 = y - h_0 u$

$\dot{x}_1 = x_2 + h_1 u$

$\dot{x}_2 = x_3 + h_2 u$

$\vdots$

$\dot{x}_{n-1} = x_n + h_{n-1} u$

$\dot{x}_n = -a_n x_1 - \dots - a_1 x_n + h_n u$

$h_0 = b_0$

$h_1 = b_1 - a_1 b_0$

$h_2 = (b_2 - a_2 b_0) + a_1 h_1$

$\vdots$

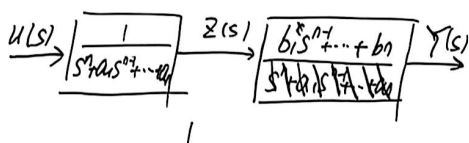


方法二:

$$G(s) = \frac{b_0 s^n + b_1 s^{n-1} + \dots + b_n}{s^n + a_1 s^{n-1} + \dots + a_n}$$

$$= b_0 + \frac{(b_1 - a_1 b_0) s^{n-1} + \dots + (b_n - a_n b_0)}{s^n + a_1 s^{n-1} + \dots + a_n}$$

① $b_0 = 0$



$$Z(s) = \frac{1}{s^n + a_1 s^{n-1} + \dots + a_n} U(s)$$

$$z^{(n)}(t) + a_1 z^{(n-1)} + \dots + a_n z(t) = u(t)$$

取 $x_1(t) = z(t)$

$$x_2(t) = \dot{x}_1(t) = \dot{z}(t)$$

$\vdots$

$$x_n(t) = \dot{x}_{n-1}(t) = z^{(n-1)}(t)$$

$$\dot{x}_n(t) = z^{(n)}(t) = -a_n x_1 - \dots - a_1 x_n + u(t)$$

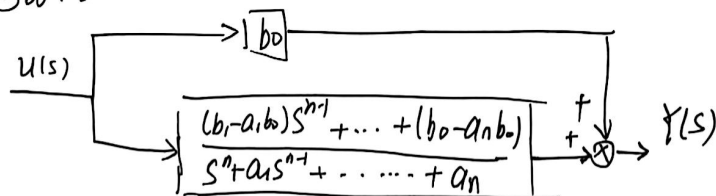
$$Y(s) = (b_1 s^{n-1} + \dots + b_n) Z(s)$$

$$y(t) = b_1 \dot{z}(t) + \dots + b_n z(t)$$

$$\text{令 } x(t) = \begin{bmatrix} x_1(t) \\ \vdots \\ x_n(t) \end{bmatrix}$$

$$\begin{cases} \dot{x}(t) = \begin{bmatrix} 0 & 1 & 0 & \dots & 0 \\ 0 & 0 & 1 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & 1 \\ -a_n & \dots & \dots & \dots & -a_1 \end{bmatrix} x(t) + \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix} u(t) \\ y(t) = [b_n \dots b_1] x(t) \end{cases}$$

② $b_0 \neq 0$



$$\begin{cases} \dot{x}(t) = \begin{bmatrix} 0 & 1 & 0 & \dots & 0 \\ 0 & 0 & 1 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & 1 \\ -a_n & -a_{n-1} & \dots & \dots & -a_1 \end{bmatrix} x(t) + \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix} u(t) \\ y(t) = [b_n - a_n b_0 \quad \dots \quad b_1 - a_1 b_0] x(t) + b_0 u(t) \end{cases}$$

例: $\ddot{y} + 5\dot{y} + 2y = 2u + \dot{u}$

$$Z(s) = \frac{1}{s^2 + 5s + 2} U(s)$$

$$\ddot{z}(t) + 5\dot{z}(t) + 2z(t) = u(t)$$

取 $x_1(t) = z(t)$

$$x_2(t) = \dot{x}_1(t) = \dot{z}(t)$$

$$x_3(t) = \dot{x}_2(t) = \ddot{z}(t)$$

$$\dot{x}_3(t) = \ddot{z}(t) = -5\dot{z}(t) - 2z(t) + u(t) = -5x_2(t) - 2x_1(t) + u(t)$$

$$Y(s) = (s + 2) Z(s)$$

$$y(t) = \dot{z}(t) + 2z(t) = x_2(t) + 2x_1(t)$$

$$\text{令 } x(t) = \begin{bmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \end{bmatrix}$$

$$\begin{cases} \dot{x}(t) = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -2 & -5 & 0 \end{bmatrix} x(t) + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u(t) \\ y(t) = [2 \quad 1 \quad 0] x(t) \end{cases}$$

法三

三、根据传递函数极点极点建立状态空间表达式

1. 单极点

$$\frac{Y(s)}{U(s)} = \frac{C_1}{s-s_1} + \frac{C_2}{s-s_2} + \dots + \frac{C_n}{s-s_n}$$

$$\text{取 } x_1(s) = \frac{1}{s-s_1} U(s)$$

$$x_2(s) = \frac{1}{s-s_2} U(s)$$

$$\vdots$$

$$x_n(s) = \frac{1}{s-s_n} U(s)$$

$$s x_1(s) - s_1 x_1(s) = U(s)$$

$\vdots$

$$s x_n(s) - s_n x_n(s) = U(s)$$

$$\dot{x}_1(t) = s_1 x_1(t) + u(t)$$

$$\dot{x}_2(t) = s_2 x_2(t) + u(t)$$

$$\vdots$$

$$\dot{x}_n(t) = s_n x_n(t) + u(t)$$

$$Y(s) = C_1 x_1(s) + \dots + C_n x_n(s)$$

$$y(t) = C_1 x_1(t) + \dots + C_n x_n(t)$$



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$$\chi(t) = \begin{bmatrix} \chi_1(t) \\ \chi_2(t) \\ \vdots \\ \chi_n(t) \end{bmatrix}$$

$$\dot{\chi}(t) = \begin{bmatrix} s_1 & 1 & & \\ & s_1 & \ddots & \\ & & \ddots & 1 \\ & & & s_1 \end{bmatrix} \chi(t) + \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{bmatrix} u(t)$$

$$y(t) = [C_1 \ C_2 \ \dots \ C_n] \chi(t)$$

$$Y(s) = C_1 \chi_1(s) + C_2 \chi_2(s) + \dots + C_n \chi_n(s)$$

$$\begin{cases} \dot{\chi}(t) = \begin{bmatrix} s_1 & & \\ & s_2 & \\ & & \ddots \\ & & & s_n \end{bmatrix} \chi(t) + \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix} u(t) \\ y(t) = [C_1 \ C_2 \ \dots \ C_n] \chi(t) \end{cases}$$

$$\text{例 8.3} \quad \frac{Y(s)}{U(s)} = \frac{2s^2 + 5s + 1}{(s-2)^3}$$

$$= \frac{19}{(s-2)^3} + \frac{13}{(s-2)^2} + \frac{2}{s-2}$$

$$\text{例 8.2} \quad \frac{\chi_2(s)}{U(s)} = \frac{2s+1}{s^3+7s^2+14s+8} = \frac{-\frac{1}{3}}{s+1} + \frac{\frac{2}{3}}{s+2} + \frac{-\frac{7}{6}}{s+4}$$

$$\chi_1(s) = \frac{1}{s+1} u(s) \quad s\chi_1(s) + \chi_1(s) = u(s)$$

$$\chi_2(s) = \frac{1}{s+2} u(s) \quad s\chi_2(s) + 2\chi_2(s) = u(s)$$

$$\vdots$$

$$\chi_3(s) = \frac{1}{s+4} u(s)$$

$$\chi(t) = \begin{bmatrix} \chi_1(t) \\ \chi_2(t) \\ \chi_3(t) \end{bmatrix}$$

$$\begin{cases} \dot{\chi}(t) = \begin{bmatrix} -1 & & \\ & -2 & \\ & & -4 \end{bmatrix} \chi(t) + \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} u(t) \\ y(t) = \begin{bmatrix} -\frac{1}{3} & \frac{2}{3} & -\frac{7}{6} \end{bmatrix} \chi(t) \end{cases}$$

2. 单重实极点

$$\frac{Y(s)}{U(s)} = \frac{*}{(s-s_1)^n} = \frac{C_1}{(s-s_1)^n} + \frac{C_2}{(s-s_1)^{n-1}} + \dots + \frac{C_n}{s-s_1}$$

$$\begin{cases} \chi_1(s) = \frac{1}{(s-s_1)^n} u(s) = \frac{1}{s-s_1} \chi_2(s) \\ \chi_2(s) = \frac{1}{(s-s_1)^{n-1}} u(s) = \frac{1}{s-s_1} \chi_3(s) \\ \vdots \\ \chi_n(s) = \frac{1}{s-s_1} u(s) \end{cases}$$

$$s\chi_1(s) - s_1\chi_1(s) = \chi_2(s)$$

$$\begin{cases} \dot{\chi}_1(t) = s_1\chi_1(t) + \chi_2(t) \\ \dot{\chi}_2(t) = s_1\chi_2(t) + \chi_3(t) \\ \vdots \\ \dot{\chi}_{n-1}(t) = s_1\chi_{n-1}(t) + \chi_n(t) \\ \dot{\chi}_n(t) = s_1\chi_n(t) + u(t) \end{cases}$$

8.3 由状态空间表达式求传递函数

$$\dot{\chi}(t) = A\chi(t) + Bu(t)$$

$$y(t) = C\chi(t) + Du(t)$$

$$s\chi(s) = A\chi(s) + Bu(s)$$

$$(sI - A)\chi(s) = Bu(s)$$

$$\chi(s) = (sI - A)^{-1}Bu(s)$$

$$Y(s) = C\chi(s) + Du(s)$$

$$= [C(sI - A)^{-1}B + D]u(s)$$

$$\therefore G(s) = C(sI - A)^{-1}B + D = \begin{bmatrix} A & B \\ C & D \end{bmatrix}$$



8.4 线性定常系统状态方程的解

一、齐次状态方程的解 $\mathcal{L}^{-1}[(sI-A)^{-1}] = e^{At}$

$$\dot{x}(t) = Ax(t)$$

$$sX(s) - x(0) = AX(s)$$

$$(sI-A)X(s) = x(0)$$

$$X(s) = (sI-A)^{-1}x(0)$$

$$\mathcal{L}^{-1}: X(t) = \mathcal{L}^{-1}[(sI-A)^{-1}]x(0)$$

$$(sI-A)\left(\frac{I}{s} + \frac{A}{s^2} + \dots + \frac{A^k}{s^{k+1}} + \dots\right)$$

$$= I + \frac{A}{s} + \dots + \frac{A^k}{s^k} + \dots$$

$$-\left(\frac{A}{s} + \dots + \frac{A^k}{s^k} + \dots\right)$$

$$= I$$

$$(sI-A)^{-1} = \frac{I}{s} + \frac{A}{s^2} + \dots + \frac{A^k}{s^{k+1}} + \dots$$

e^{At} 矩阵指数

$$x(t) = e^{At}x(0)$$

例 8.4 $\dot{x} = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix} x(t)$

$$x(0) = \begin{bmatrix} x_1(0) \\ x_2(0) \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

求 $x(t)$

$$(sI-A)^{-1} = \begin{bmatrix} s & -1 \\ 2 & s+3 \end{bmatrix}^{-1}$$

$$= \frac{\begin{bmatrix} s+3 & 1 \\ -2 & s \end{bmatrix}}{s^2+3s+2}$$

$$= \begin{bmatrix} \frac{s+3}{(s+2)(s+1)} & \frac{1}{(s+2)(s+1)} \\ \frac{-2}{(s+2)(s+1)} & \frac{s}{(s+2)(s+1)} \end{bmatrix}$$

$$= \begin{bmatrix} \frac{2}{s+1} - \frac{1}{s+2} & \frac{1}{s+1} - \frac{1}{s+2} \\ \frac{-2}{s+1} + \frac{2}{s+2} & \frac{-1}{s+1} + \frac{2}{s+2} \end{bmatrix}$$

$$e^{At} = \mathcal{L}^{-1}[(sI-A)^{-1}] = \begin{bmatrix} 2e^{-t}-e^{-2t} & e^{-t}-e^{-2t} \\ -2e^{-t}+2e^{-2t} & -e^{-t}+2e^{-2t} \end{bmatrix}$$

$$x(t) = e^{At}x(0)$$

$$= \begin{bmatrix} 2e^{-t}-e^{-2t} \\ -2e^{-t}+2e^{-2t} \end{bmatrix}$$

二、矩阵指数的性质

$$\textcircled{1} \frac{d}{dt} e^{At} = A e^{At} = e^{At} A$$

$$\textcircled{2} e^{At}|_{t=0} = I$$

$$\textcircled{3} (e^{At})^{-1} = e^{-At}$$

$$\textcircled{4} e^{A(t_1+t_2)} = e^{At_1} \cdot e^{At_2}$$

$$\textcircled{5} n \text{ 为整数}, (e^{At})^n = e^{nAt}$$

$$\textcircled{6} \text{ 若 } AB=BA, e^{(A+B)t} = e^{At} \cdot e^{Bt}$$

$$\textcircled{7} P \text{ 非奇异}, e^{P^T A P t} = P^{-1} e^{At} P$$

三、非齐次方程的解

$$\dot{x}(t) = Ax(t) + Bu(t)$$

1. 一般性

$$\dot{x}(t) - Ax(t) = Bu(t)$$

$$e^{-At} [\dot{x}(t) - Ax(t)] = e^{-At} Bu(t)$$

$$\int_0^t \frac{d}{dt} [e^{-At} x(t)] dt = \int_0^t e^{-At} Bu(t) dt$$

$$[e^{-At} x(t)] \Big|_0^t = \int_0^t e^{-At} Bu(t) dt$$

$$e^{-At} x(t) - x(0) = \int_0^t e^{-At} Bu(t) dt$$

$$x(t) = e^{At} x(0) + \int_0^t e^{A(t-\tau)} Bu(\tau) d\tau$$

若 $t_0 \neq 0$

$$x(t) = e^{At} x(t_0) + \int_{t_0}^t e^{A(t-\tau)} Bu(\tau) d\tau$$



2. 拉氏变换法

$$\dot{x}(t) = Ax(t) + Bu(t)$$

$$sX(s) - x(0) = AX(s) + BU(s)$$

$$(sI - A)X(s) = x(0) + BU(s)$$

$$X(s) = (sI - A)^{-1}x(0) + (sI - A)^{-1}BU(s)$$

$$\mathcal{L}^{-1}: x(t) = \mathcal{L}^{-1}[(sI - A)^{-1}x(0)] + \mathcal{L}^{-1}[(sI - A)^{-1}BU(s)]$$

例 8.5

$$\dot{x}(t) = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix} x(t) + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u(t)$$

$$u(t) = 1(t) \quad x(0) = x_0$$

解: $(sI - A)^{-1}BU(s) = \dots$

$$\int_0^t e^{A(t-\tau)} Bu(\tau) d\tau = \dots$$

$$x(t) = e^{At}x(0) + \int_0^t e^{A(t-\tau)} Bu(\tau) d\tau$$

8.7 线性系统的可控性与可观测性

$$\begin{cases} \dot{x}_1 = -\frac{1}{R_1 C_1} x_1 + \frac{1}{R_1 C_1} u_1 \\ \dot{x}_2 = -\frac{1}{R_2 C_2} x_2 + \frac{1}{R_2 C_2} u_2 \end{cases}$$

一. 可控性与可控性判据

$$\begin{cases} \dot{x}(t) = Ax(t) + Bu(t) \\ y(t) = Cx(t) + Du(t) \end{cases} \quad x \in \mathbb{R}^n$$

可控性矩阵

$$Q_k = [B \quad AB \quad A^2B \quad \dots \quad A^{n-1}B]$$

充要条件: $\text{rank } Q_k = n$

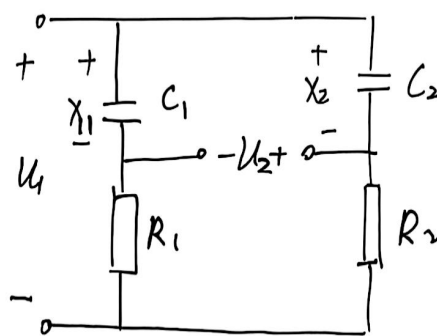
例 8.6 $\dot{x} = \begin{bmatrix} -4 & 1 \\ 2 & -3 \end{bmatrix} x + \begin{bmatrix} 1 \\ 2 \end{bmatrix} u$

$$Q_k = \begin{bmatrix} 1 & -2 \\ 2 & -4 \end{bmatrix}$$

例 8.7 $\dot{x} = \begin{bmatrix} -2 & 1 \\ 0 & -3 \end{bmatrix} x + \begin{bmatrix} 0 & 1 \\ 1 & -1 \end{bmatrix} u$

$$Q_k = \begin{bmatrix} 0 & -1 & 1 & -3 \\ 1 & 1 & -3 & 3 \end{bmatrix}$$

例 8.8



u_1 输入 u_2 输出

求不可控、不可观条件

$$\dot{x} = \begin{bmatrix} -\frac{1}{R_1 C_1} & 0 \\ 0 & -\frac{1}{R_2 C_2} \end{bmatrix} x + \begin{bmatrix} \frac{1}{R_1 C_1} \\ \frac{1}{R_2 C_2} \end{bmatrix} u$$

$$y = [1 \quad -1] x$$

$$Q_g = \begin{bmatrix} 1 & -1 \\ -\frac{1}{R_1 C_1} & \frac{1}{R_2 C_2} \end{bmatrix}$$

二. 可观性及其判据

可观性矩阵

$$Q_g = \begin{bmatrix} C \\ CA \\ \vdots \\ CA^{n-1} \end{bmatrix}$$

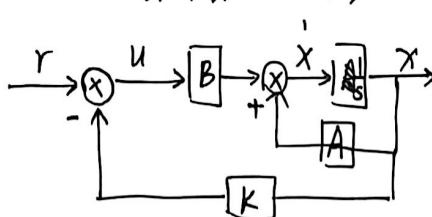
充要条件: $\text{rank } Q_g = n$

8.8 线性系统状态反馈与极点位置

$$\dot{x} = Ax + Bu$$

条件: 可控

$$\text{引入 } u = r - Kx$$



$$\dot{x} = Ax + B(r - Kx) = (A - BK)x + Br$$



$$\text{例 8.9} \quad \begin{cases} \dot{x} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & -2 & -3 \end{bmatrix} x + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u \\ y = [1 \ 0 \ 0] x \end{cases}$$

使闭环极点 $\lambda_1 = -5, \lambda_{2,3} = -2 \pm j2$

解: 设 $k = [k_1 \ k_2 \ k_3]$

$$\begin{aligned} Q_k &= [B \ AB \ A^2B] \\ &= \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & -3 \\ 1 & -3 & 7 \end{bmatrix} \end{aligned}$$

$\because \text{rank } Q_k = 3 \quad \therefore \text{可控}$

$$\begin{aligned} |sI - (A - Bk)| &= \begin{vmatrix} s & -1 & 0 \\ 0 & s & -1 \\ k_1 & 2+k_2 & s+3+k_3 \end{vmatrix} \\ &= s^3 + (k_3+3)s^2 + (k_2+2)s + k_1 \\ &= (s+5)(s+2+j2)(s+2-j2) \\ &= s^3 + 9s^2 + 28s + 40 \end{aligned}$$

$$k_3 = 6 \quad k_2 = 26 \quad k_1 = 40$$

$$\therefore k = [40 \ 26 \ 6]$$

